



IoT-driven dynamic replenishment of fresh produce in the presence of seasonal variations: A deep reinforcement learning approach using reward shaping[☆]

Zihao Wang^{a,*,}, Wenlong Wang^{a,*}, Tianjun Liu^a, Jasmine Chang^b, Jim Shi^b

^a College of Economics & Management, Northwest A&F University, Yangling, China

^b Tuchman School of Management, New Jersey Institute of Technology, Newark, NJ, USA

ARTICLE INFO

Editor: Dr J Zhu

Keywords:

Dynamic replenishment of fresh produce
Seasonal variations

IoT

DRL

Reward shaping

ABSTRACT

Internet of things (IoT) has been transforming inventory management disruptively by linking and synchronizing inventory products together. It is one of the driving forces for the prevailing innovation of AgriTech. For fresh produce replenishment in the presence of its inherent seasonal variations, not only can IoT devices capture bidirectional seasonal information of lead time and demand, but also detect fresh produce loss and waste (FPLW) caused by deterioration. With the aid of the massive data collected by IoT, we propose a data-driven deep reinforcement learning (DRL) approach using reward shaping, called DQN-SV-RS, to optimize the dynamic replenishment policy for a fresh produce wholesaler, specifically addressing the challenge posed by seasonal variations. Experimental results show that our DQN-SV-RS approach yields significant improvements for fresh produce supply chain (FPSC) inventory management, especially achieving a remarkable reduction in FPLW. As a core innovation in our DQN-SV-RS approach, the introduced reward shaping can significantly mitigate lost sales and inventory holding, thereby lowering the total cost. Furthermore, with numerical experiments based on real business data, our proposed approach is demonstrated with plausible robustness and scalable applicability.

1. Introduction

As the world's population is projected to reach 9.7 billion by 2050, agriculture needs to become more productive and sustainable than ever [1]. To this end, agricultural technology (AgriTech), leveraging a variety of state-of-the-art technologies, has witnessed significant innovation, widespread implementation, and burgeoning development. One of the most disruptive technologies in AgriTech is the Internet of Things (IoT). IoT is a network of interrelated devices (e.g., sensors) that connect and exchange data with other devices and the cloud. For example, IoT-enabled smart farming leverages sensors and gateways to collect and analyze data. Furthermore, IoT has ushered in a new era for inventory management [2]. Smart IoT sensors can continuously monitor inventory, track products in real-time, and alert management when items are running low and need replenishment. For example, Walmart stores, supported by the IoT, have achieved a 63 % increase in replenishment efficiency and a 16 % reduction in out-of-stock occurrences. Similarly,

Decathlon has utilized IoT to speed up inventory inspections in logistics centers and warehouses. Despite these advancements, replenishment management in the fresh produce supply chain (FPSC) presents unique challenges owing to the seasonality of fresh produce, which causes variations in supply and demand [3]. The IoT's ability to enhance data quality, freshness, and availability offers the potential for improving decision-making efficiency. Consequently, addressing this challenge necessitates the development of efficient dynamic replenishment for fresh produce, which can be facilitated by IoT technology.

Historically, inventory management problems have primarily been addressed by individual companies or single supply chain entity [4–6]. However, in the context of FPSCs, wholesalers typically serve as a pivotal middleman, bridging upstream producers and downstream retailers to coordinate resources between supply and demand [7]. Therefore, wholesalers' replenishment strategies should not disregard interactions with other FPSC entities, particularly during seasonal variations, which can significantly impact their operations at large. Given

[☆] Area: Data-Driven Analytics This manuscript was processed by Associate Editor Xiang Fang

^{*} Corresponding author.

E-mail addresses: zihao.wang@nwafu.edu.cn (Z. Wang), wl.wang@nwafu.edu.cn (W. Wang), ltj168168@nwsuaf.edu.cn (T. Liu), jschang@njit.edu (J. Chang), jshi@njit.edu (J. Shi).

<https://doi.org/10.1016/j.omega.2025.103299>

Received 12 August 2024; Accepted 6 February 2025

Available online 12 February 2025

0305-0483/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

the superiority of IoT in monitoring deterioration and collecting information on seasonal variations, it is promising for wholesalers to leverage this emerging powerful technology to accomplish dynamic replenishment for fresh produce.

Although a growing body of research is studying the replenishing of fresh produce, scant attention has been devoted to addressing the chronic challenge of seasonal variations. Seasonal variation is the inherent feature of fresh produce, and it describes that the lead time and demand vary seasonally between peaks and troughs. If seasonality is not considered while making replenishment decisions, wholesalers will be more likely to suffer from unexpected shortages of inventory in the season, as well as overage of inventory causing fresh produce loss and waste (FPLW) during the off-season. Typically, most of extant literature (if not all) assumes that the lead time is either zero or a fixed positive value [8,9], whereas demand is always treated as a random variable following a known probability distribution [6,10] or linear time-dependent factor [11,12]. However, such an assumption sounds overly idealistic. In the presence of seasonal variations, these traditional replenishment approaches become infeasible to derive the optimality of closed-form expressions [4]. In terms of fresh produce replenishment, most research focuses on storage [13], routing problems [14,15], and fresh-based price reduction [16], but seasonal variations are not well discussed or documented, even though the circulation and trade of fresh produce are dramatically affected by uncertainties pertaining to seasonality.

In this study, we aim to bridge the aforementioned research gap by integrating seasonal variations into the dynamic replenishment for an FPSC wholesaler utilizing IoT. To achieve this goal, two challenges must be overcome: (1) How can we factor in the information of seasonal variations that are real-timely collected by IoT devices? (2) With the massive amount of information collected by IoT devices, how can we obtain the optimal replenishment strategy? To this end, we first employ cosine functions with stochastic disturbances to model the seasonal variations in the lead time and demand. Next, we set up a Markov decision process (MDP) for dynamic replenishment in the presence of seasonal variations. Given the substantial state information provided by IoT devices, we develop a data-driven deep reinforcement learning (DRL) approach with reward shaping, called DQN-SV-RS, to address this dynamic replenishment problem. In particular, the proposed DQN-SV-RS approach is based on the deep Q-network (DQN) algorithm. In contrast with the standard approaches, such as dynamic programming (DP) or reinforcement learning methods in "tabular form", the DQN can efficiently address the curse of dimensionality from which the standard approaches have been suffering [17]. Experimental results show that, bolstered by reward shaping, the performance of the DQN-SV-RS significantly outperforms a wide range of benchmarks, including Rolling BSP-low-EW (Rolling-BLE) studied by Haijema and Minner [6], Predict-then-optimize (PTO) by Elmachtoub and Grigas [18], End-to-End (E2E) by Qi et al. [19], and DQN by De Moor et al. [5] and Oroojlooyjadid et al. [20]. Moreover, we demonstrate the effectiveness of the reward shaping through a two-way fixed effects model. An empirical study with real-world data further supports the efficiency of DQN-SV-RS in practical scenarios, whereas robustness analyses highlight its generality in FPSCs. To summarize, the main contributions of this paper include:

- 1) A novel framework is proposed to handle massive amounts of IoT-collected information for the dynamic replenishment of fresh produce (cf. Section 3).
- 2) The bidirectional information on seasonal variations in lead time and demand are integrated into the dynamic replenishment problem for FPSCs (cf. Section 4).
- 3) Reward shaping is developed to enhance a DQN-based dynamic replenishment approach with mathematical derivation (cf. Section 5).

- 4) Comprehensive experiments are conducted to demonstrate the superior performance against various commonly used benchmarks (cf. Section 6).

2. Literature review

2.1. Nonseasonal fresh produce replenishment

In a pioneering work, Bai and Kendall [21] first separated the replenishment of fresh produce from general perishables due to its continuous deterioration, sparking further research on fresh produce replenishment. Qin et al. [22] addressed pricing and inventory control with concurrent quality and quantity deterioration, whereas Hou et al. [7] studied the replenishment policy for a fresh produce wholesaler, considering cold storage and disposal. Qiao et al. [16] proposed a joint ordering and price reduction policy with competitive price- and freshness-based demand. However, these studies focused primarily on freshness or deterioration, with limited attention given to the direct effects of time-dependent demand and lead time on replenishment policies.

For time-dependent demand, Macías-López et al. [23] and Sivashankari and Ramachandran [24] developed perishable inventory models that incorporate various time-dependent demand functions without considering the relationship between perishable properties and demand. Xie et al. [12] and Kumar et al. [11] approached the perishable nature from different angles: the former assumed linear demand reduction in terms of shelf life, whereas the latter assumed that demand increases as the harvest season approaches. Nevertheless, seasonality—a ubiquitous characteristic of fresh produce demand—has been largely overlooked. To the best of our knowledge, only Arnade et al. [25] and Udenio et al. [26] had modeled seasonal time-varying demand, but they focused on nonperishable items.

Concerning lead time, most perishable inventory studies treat lead time as a constant or negligible [8,9], with a few exceptions considering lead time as stochastic, with a discrete distribution [27], or subject to transportation disruptions [28]. However, lead time can vary seasonally due to changes in procurement locations during different production seasons [29], especially for fresh produce. For example, in China, durians are supplied from southern regions during the spring and summer seasons, whereas they are imported from Southeast Asia during the autumn and winter seasons.

Based on the discussion above, the bidirectional seasonal variations in both demand and lead time—critical factors in the fresh produce supply chain—should be considered simultaneously when making replenishment decisions. This becomes more feasible than ever thanks to data enrichment and availability collected by IoT.

2.2. DRL in agricultural supply chain

DRL is revolutionizing the agricultural sector. Alibabaei et al. [30] developed a DQN to optimize irrigation for tomato farmers by using soil water content and evapotranspiration data. Din et al. [31] employed a double deep Q-network (DDQN) with a gridded farm map for monitoring crop health. Moreover, DRL has also shown potential in inventory management, with an asynchronous advantage actor-critic (A3C) improving inventory management performance [32], applying the DQN solving the problem of beer game [20] and addressing perishable inventory management [5]. In addition, Deep learning (DL) was employed to establish an end-to-end (E2E) framework for the replenishment task [19].

In previous studies that applied DRL or DL to inventory management, demand was typically assumed to follow a random distribution, and the lead time was fixed as a constant or zero. While Qi et al. [19] utilized real business data, both demand and lead times remained random. In contrast, our study considers the unique characteristics of the FPSC, where both demand and lead times exhibit seasonal randomness.

Practically, the consideration of seasonal randomness will largely increase the dimensionality in the state space, which exacerbates the curse of dimensionality. Therefore, one of the key contributions of this paper is to leverage DRL to solve the dynamic replenishment problem for fresh produce, considering the bidirectional seasonal variations in demand and lead time. Moreover, to the best of our knowledge, only Yang et al. [33] had explored the agricultural supply chain, using proximal policy optimization (PPO) to optimize pricing and information strategies for fresh produce retailers. Given the bidirectional seasonal variations, traditional policies such as BSP-low-EW [6] become suboptimal because of the infeasibility to derive closed-form expressions [4]. Although dynamic programming can offer assistance, the curse of dimensionality stands out as the state space grows. Given the ability of DRL to handle the issue of multidimensional states, we employ a DQN and incorporate a seasonal state to address fresh produce replenishment when seasonal variations are bidirectional.

2.3. IoT in inventory management

Real-time seamless connectivity between physical systems and digital information has revolutionized management and operations, particularly in FPSCs [33]. In the context of smart farming, farmers can monitor crop growth and anomalies to effectively prevent diseases or infestations that can harm yield. Arable¹ and Semios² can serve as good examples of how this use case can be applied in real life.

IoT technology has been praised in terms of generating rich data for inventory management and thus optimizing inventory costs and improving other performances. On the one hand, the IoT can continuously collect real-time information (e.g., inventory levels, demands, and lead times) to facilitate fully automated inventory review. For example, an RFID-enabled inventory system makes product transfer and review more transparent, allowing supply chain members to selectively use and share this information [34], which helps improve overall product replenishment [35]. On the other hand, the IoT can be used to monitor fresh produce quality, thereby reducing FPLW.

In the literature, several sensing technologies that detect the physical or chemical properties of fresh produce—such as pH [36], time–temperature [37]—can be used to monitor the quality of fresh produce. Ubiquitous sensors and smart devices further contribute to a wealth of information about inventory freshness, allowing decision-makers to detect and monitor waste. There is further evidence underscoring the innovative impact of the IoT on perishable inventory management. For example, Hassini et al. [38] argued that deploying the IoT in food supply chains can reduce food waste, enhance decision-making via IoT data, and address various operational challenges, particularly in food inventory management. This, in turn, fosters the resilience and sustainability of food supply chains. Özbilge et al. [39] developed a quality-dependent newsvendor model and leveraged IoT-enabled state-tracking technologies to accurately estimate the quality of fresh goods and their impact on consumer demand. However, the lack of a platform that can integrate data from different devices prevents the full exploitation of the data to improve the performance of the whole supply chain [40]. IoT can significantly enhance data quality, freshness and availability, leading to improved decision-making efficiency. In this sense, we use the IoT to review inventories, observe seasonal variations, and detect FPLW for dynamic replenishment via DRL.

3. System framework

In this section, we characterize a system framework for fresh produce replenishment driven by IoT technology, which operates within a dynamic replenishment model. By leveraging the IoT infrastructure, we

focus on a wholesaler, as a focal supply chain intermediary, to optimize replenishment in the face of seasonal variations, minimize total cost, and reduce FPLW.

As depicted in Fig. 1, this system consists of three key components: (1) the wholesaler's warehouse, which is monitored by IoT devices that upload data to the IoT cloud server (blue block); (2) the IoT cloud server, which processes and stores the data; and (3) the DQN agent, which is driven by data from the IoT cloud server to dynamically replenish fresh produce (purple block).

In operation, IoT devices in the blue block first collect real-time data across various dimensions, such as seasonal variations, inventory levels, and product quality. For data related to seasonal variations and inventory, RFID readers automatically capture tags affixed to the packaging of fresh produce during receiving, issuing, and storing [38,41]. Novel RFID systems have been shown to increase inventory efficiency and reduce identification times by >50 % [42], thus eliminating the need for traditional manual inventory counting. The data are then uploaded to the IoT cloud server, which typically contains the inbound and outbound quantities, along with inventory levels categorized by age group, freshness, quality, timestamps, etc. For the quality data, a series of IoT devices monitor fresh produce inventory in real time, tracking parameters such as temperature, humidity, gas composition, visual appearance, etc. [36,37]. Once the data are received, the IoT cloud server computes the demand, lead time, FPLW, and net inventory, providing valuable insights into inventory, ordering, and seasonal variations. Finally, the DQN agent in the purple block assesses these data to learn and choose the optimal actions for dynamic replenishment, whereas the IoT cloud server computes the costs and rewards for the agent.

4. Dynamic replenishment of fresh produce considering seasonal variations

This study considers a periodic review supply chain for a type of fresh produce, where a wholesaler fulfills the demand from a retailer. The produce experiences seasonal variations in demand and lead time, as well as time-varying deterioration, with a maximum shelf life of k periods (e.g., days). The wholesaler receives orders on the basis of the retailer's needs rather than direct market demand, and the wholesaler's inventory affects the retailer's orders, a factor often overlooked in replenishment models [5,32,43]. The main notations in this formulation are summarized in Table 1.

To facilitate scenario description and dynamic replenishment, the following assumptions are made in line with existing practices and the literature:

- (1) The wholesaler adopts dynamic replenishment, whereas the retailer follows a traditional replenishment policy.
- (2) The wholesaler has sufficient supplies from different producers in different seasons, but the current inventory poses an upper limit for its retailer's order.
- (3) Storage capacities are uncapped and nonnegative for both the wholesaler and the retailer.
- (4) The retailer is willing to share market demand information with the wholesaler via IoT in a cooperative arrangement.
- (5) Owing to the lead time and one inventory review period, orders are not received immediately. In addition, the times of receiving, issuing and ordering are disregarded.
- (6) The fulfilling policy follows the first-in-first-out (FIFO) principle.

4.1. Process of dynamic replenishment under seasonal variations

We describe our dynamic replenishment as the following four-stage sequence of replenishment events [44,45], as shown in Fig. 2.

¹ <https://www.arable.com/>.

² <https://semios.com/>.

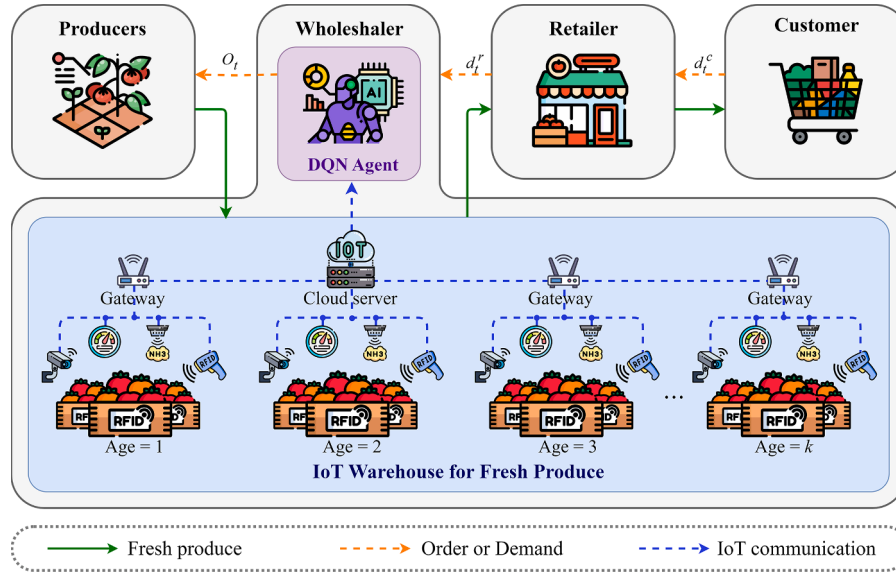


Fig. 1. System Framework: IoT-driven Dynamic Replenishment of Fresh Produce.

Table 1

The main replenishment notations with explanation.

Notation	Definition and Explanation
d_t	Downstream demand at period t .
l_t	Lead time at period t .
$i_{t,j}$	Inventory level at period t , which was received in period j , for $t \geq j$.
$\tilde{i}_{t,j}$	Remaining inventory after $i_{t,j}$ satisfied d_t .
$w_{t,j}$	FPLW produced by perishing $\tilde{i}_{t,j}$.
$ni_{t,j}$	Net inventory remaining after $\tilde{i}_{t,j}$ perished.
O_t	Order (replenishment quantity) at period t .
R_t	Receipt of the fresh produce at period t .
NI_t	Total amount of net inventories at period t .
LS_t	Total amount of the shortages at period t .
W_t	Total amount of the FPLWs at period t .
c_o	Unit ordering cost.
c_h	Unit holding cost.
p_l	Unit lost sales penalty.
p_w	Unit waste penalty.
k	Maximum shelf life.

(1) *Reception stage*. The fresh produce is received, and the inventory along with the lead time is recorded via RFID. At the start of each period t , the receipt R_t is received and added to the inventory with age 1. The inventory $I_t = (i_{t,t} = R_t, i_{t,t-1}, i_{t,t-2}, \dots, i_{t,t-k+1})$ and the lead time l_j of the receipt are tracked via RFID. Since the receipt corresponds to some orders O_j placed in period j , the lead time l_j of O_j is known only once O_j is received. Therefore, the relationship satisfies $j + l_j + 1 = t$ [27].

(2) *Serving stage*. The demand is fulfilled from the current inventory, and the remaining inventory is calculated by the cloud server. These data are recorded via RFID. After demand d_t is fulfilled, the

remaining inventory is updated as $\tilde{I}_t = (\tilde{i}_{t,t}, \tilde{i}_{t,t-1}, \tilde{i}_{t,t-2}, \dots, \tilde{i}_{t,t-k+1})$, where

$$\tilde{i}_{t,j} = \begin{cases} [i_{t,j} - d_t]^+, & \text{for } j = t - k + 1 \\ \left[i_{t,j} - \left[d_t - \sum_{m=t-k+1}^{j-1} i_{t,m} \right]^+ \right]^+, & \text{for } j = t - k + 2, \dots, t \end{cases} \quad (1)$$

(3) *Deterioration stage*. IoT devices monitor the remaining inventory and FPLW. This FPLW includes items that have deteriorated or exceeded their shelf life k . The FPLW is calculated via the deterioration model F with IoT data (temperature, humidity, CO_2 , etc.),

$$w_{t,j} = \int_t^{t+1} F(\text{temperature}_{t,j}, \text{humidity}_{t,j}, CO_{2,t,j}, \dots) dt, \quad (2)$$

(4) *Sourcing stage*. The net inventory is updated, an order is issued via the DQN algorithm, and the total cost is computed. The end-of-period net inventory is first updated as $ni_{t,j} = \tilde{i}_{t,j} - w_{t,j}$. The inventory for the next period is updated as $i_{t+1,j} = ni_{t,j}$, and the order O_t is placed via the IoT-driven DQN. The IoT cloud server computes the total inventory cost as follows:

$$C_t = c_o O_t + c_h NI_t + p_l LS_t + p_w W_t, \quad (3)$$

where $NI_t = \sum_{j=t-k+1}^t ni_{t,j}$, $LS_t = \left[d_t - \sum_{j=t-k+1}^t i_{t,j} \right]^+$, and $W_t = \sum_{j=t-k+1}^t w_{t,j}$.

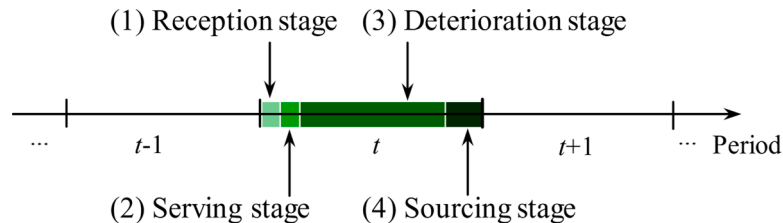


Fig. 2. Four-stage Sequence of Replenishment Events.

4.2. Seasonal variations

This study considers seasonal variations in both the upstream supply and downstream demand of the fresh produce. On the upstream side, lead times of supply vary seasonally due to the different geographical outlets of multiple producers and the limited supply seasons from producers [29]. For example, U.S. blueberry production is geographically and seasonally dependent, with harvests in the southeastern United States from March to June and in the Pacific Northwest from June to December [46]. As transportation speeds are relatively faster during harvest seasons in local areas but become slower at other times and in distant regions, it results seasonal variations in lead time.

On the downstream side, the randomness in fresh produce demand has traditionally been driven by the seasonal availability of fresh produce. According to the National Food Survey in Great Britain [47], vegetable consumption was nearly three times greater from April to September than from October to March, while total fruit consumption increased by 12 %. Despite technological advancements allowing year-round supply (e.g., vertical farming), demand still follows seasonal trends driven by environmental, physiological, and psychological factors, particularly for agricultural commodities [48]. Therefore, the demand for fresh produce varies seasonally.

In the literature, Arnade et al. [25] used seasonal trigonometric variables to construct a demand model for perishable agricultural products. Mathematically, we adopt a similar approach, modeling seasonal demand as a cosine function; cf. Udenio et al. [26] and Chan et al. [49]:

$$d_t = \left\lfloor \frac{d_{\max} + d_{\min}}{2} - \frac{d_{\max} - d_{\min}}{2} \cos\left(\frac{4\pi t}{T}\right) + \eta_t \right\rfloor, \quad (4)$$

where $\lfloor \cdot \rfloor$ is the floor operation, $d_{\min}$ and $d_{\max}$ are the minimum and maximum demand, respectively, T is the episode length (i.e., T periods), and η_t is random noise distributed as $\eta_t \sim N(0, 1)$. This simplified demand function captures seasonal variations in fresh produce, facilitating the analysis and experiments in this study. This approach also allows us to test the robustness of our DRL model under varying seasonal strengths. For example, in Section 6.4.1, we show that stronger seasonality improves the efficiency of our DRL approach.

In a similar vein to the demand function, the lead time is modeled as follows:

$$l_t = \left\lfloor \frac{l_{\max} + l_{\min}}{2} + \frac{l_{\max} - l_{\min}}{2} \cos\left(\frac{4\pi t}{T}\right) + \eta_t \right\rfloor, \quad (5)$$

where $l_{\max}$ and $l_{\min}$ are the minimum and maximum lead times, respectively. For illustration, Fig. 3 shows the seasonal variations generated by Eqs. (4) and (5) with $d_{\min} = 1$, $d_{\max} = 6$, $l_{\min} = 1$, $l_{\max} = 3$ and $T = 28$. The shaded areas around the curves represent the associated 95 % confidence intervals (CIs), providing a reasonable range of the true value distribution. With these cosine functions, the seasonality of agricultural commodities can be statistically characterized via our models.

5. DQN-based dynamic replenishment approach using reward shaping

This section introduces a DQN-based approach for deriving the optimal replenishment policy with the objective of minimizing the total cost and mitigating FPLW and stockouts. We first formulate an MDP for dynamic replenishment with seasonal variations and then introduce the DQN structure and the approach based on reward shaping.

5.1. Markov decision process (MDP) for dynamic replenishment

The dynamic replenishment problem is formulated as an MDP, which models a sequential decision problem for DRL agents to learn from numerous interactions with their environment. First, the agent selects an action a_t on the basis of the current state s_t . Then, the environment transitions to the next state s_{t+1} in response to the action a_t and generates a reward r_t to evaluate the effectiveness of the action. The agent aims to maximize the cumulative discounted reward, $\mathcal{R}_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}$, where the discount factor $\gamma \in (0, 1)$ determines the relative importance of future rewards. Consequently, an MDP is represented as a tuple $M = (s, a, r, \gamma)$ describing transitions within the system until a terminal state is reached. We next assess the state, action, and reward definitions used when dynamically replenishing fresh produce under seasonal variations.

5.1.1. States

The complete state of dynamic replenishment encapsulates two substates: (1) the seasonal state ss_t , and (2) the inventory position state si_t . First, benefiting from the IoT's robust ability of collecting vast amounts of data, the system captures seasonal data that include lead times and demands. Consequently, the seasonal state is defined as:

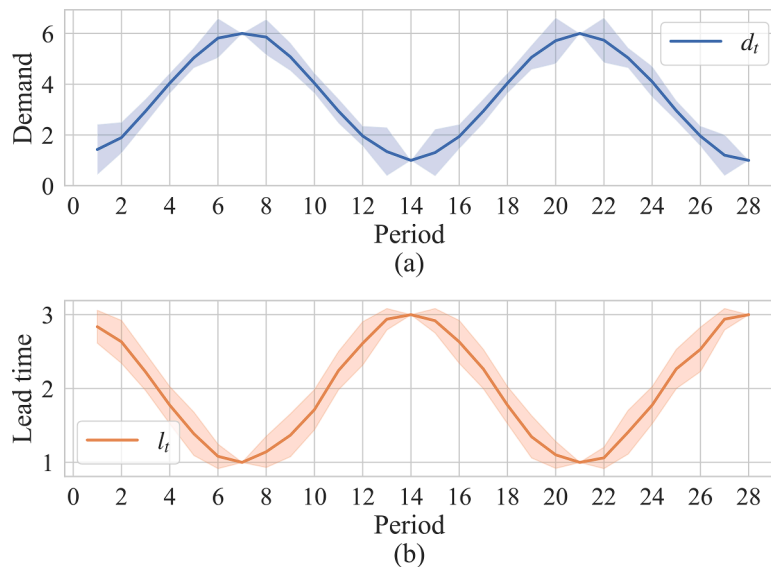


Fig. 3. Seasonal Variations: Distributions and Averages of (a) Demand and (b) Lead Times.

$$ss_t = (L_t^w, L_t^r, D_t^r, D_t^c), \forall ss_t \in \mathcal{SS}, \quad (6)$$

where $\mathcal{SS}$ represents the set of all possible seasonal states. Here, L_t^i and D_t^i refer to the historical lead time and demand data over the last τ periods for player i (retailer r , wholesaler w , or customer c).

Lead times are stochastic, meaning that they are observed only when produce is received in the current period. Consequently, the most recent observable lead time is $l_{t-\tau-1}^i$, which corresponds to l_{t-2}^i in this study setting. Additionally, demand data are only available up to period $t-1$, as downstream demand is not conveyed at the start of each period. On the basis of these observations, the historical data of lead time and demand are defined as $L_t^i = (l_{t-\tau-1}^i, l_{t-\tau+1}^i, \dots, l_{t-2}^i)$ and $D_t^i = (d_{t-\tau}^i, d_{t-\tau+1}^i, \dots, d_{t-1}^i)$. Finally, for unrarried orders with unknown lead times, their lead times are assigned a placeholder value, such as $l_j^i = -1$.

Second, following the traditional stock-age policies [6] and recent DRL applications in the literature of inventory management [5,20,32,43], the inventory position si_t is characterized by the inventory vector $I_t = (i_{t,t} = R_t, i_{t,t-1}, i_{t,t-2}, \dots, i_{t,t-k+1})$ and the cumulative outstanding order quantity Q_t (i.e., the total quantity of unreceived orders). Here, the received delivery R_t is the sum of the orders arriving in the current period. Owing to the stochastic nature of lead times, the exact lead time l_j^w becomes known only until the corresponding order is received. Accordingly, the receipt is expressed as $R_t = \sum_{j=t-p_{\max}^w-1}^{t-p_{\min}^w-1} O_j \cdot \mathbb{I}_{(j+l_j^w+1=t) \wedge (l_j^w \neq -1)}$, where $\mathbb{I}_x$ is an indicator function that equals 1 if x is true, and 0 otherwise.

The inventory position state si_t is then described as:

$$si_t = (I_t, Q_t), \forall si_t \in \mathcal{SI} \quad (7)$$

where $\mathcal{SI}$ denotes the set of all possible inventory position states and the outstanding orders. Both I_t and Q_t are retrieved from the inventory data stored in the IoT cloud server.

Finally, the complete state s_t is denoted as

$$s_t = (si_t, ss_t), \forall s_t \in \mathcal{SI} \times \mathcal{SS}. \quad (8)$$

5.1.2. Actions

The order quantity is unrestricted as any integer in $[0, +\infty)$. However, as DQN needs a finite-size action space [20], the action space is constrained to a set of feasible order quantities, denoted as $\mathcal{A}$. Accordingly, the action a_t is set as:

$$a_t = O_t, \forall a_t \in \mathcal{A}, \quad (9)$$

where $\mathcal{A} = \{0, 1, \dots, O_{\max}\}$ is a feasible action space and where $O_{\max}$ represents the maximum order quantity.

5.1.3. Rewards

At each period t , the wholesaler agent selects the optimal action a_t on the basis of the current state s_t . The effectiveness of the action is evaluated by the reward r_t , which is the negation of the total cost,

$$r_t = -C_t = -(c_o O_t + c_h N_t + p_l S_t + p_w W_t). \quad (10)$$

Our goal is to find a policy π that maps each state to an action to maximize the cumulative discounted reward $\mathcal{R}_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}$. While traditional dynamic programming approaches, such as numerical value iterations, can achieve this, they suffer from the curse of dimensionality, where computational complexity grows exponentially with the problem scale. A potential solution is DRL, such as a DQN, which reduces the computational cost by gradually converging to the optimal policy through trial-and-error, although it still faces the curse of dimensionality as the state and action spaces expand.

5.2. Deep Q-network (DQN)

A DQN integrates value function approximation and neural network techniques, employing two networks and experience replay for training the neural network [17]. Unlike the traditional Q-learning which stores Q-values $Q(s, a)$ in a table, the DQN employs two neural networks to approximate this function for large state spaces where table-based methods are impractical. The *Q-value function* is defined as the expected return from state s and action a , following a policy π :

$$Q(s, a) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} | s_0 = s, a_0 = a, \pi \right]$$

where r_{t+1} is the reward received at time $t+1$, and where $\gamma \in [0, 1]$ is the discount factor that determines the current value of future rewards. Specifically, when the DQN converges, the *optimal Q-value function* satisfies the *Bellman equation*:

$$Q^*(s_t, a_t) = \mathbb{E}_{s_{t+1}} [r(s_t, a_t) + \gamma \max_{a_{t+1} \in \mathcal{A}} Q^*(s_{t+1}, a_{t+1})].$$

In the DQN, the Q-value function is approximated via two neural networks: the main network $Q(s, a; \theta)$ and the target network $\hat{Q}(s, a; \hat{\theta})$, where θ and $\hat{\theta}$ represent the weights of the main and target networks, respectively. These two networks share the same architecture but have distinct weights, which allow for more stable training.

The main network $Q(s, a; \theta)$ is responsible for selecting actions during training. Given the current state s_t , the agent selects an action $a = \text{argmax}_a Q(s, a; \theta)$ with probability $(1 - \epsilon)$, where $0 \leq \epsilon \leq 1$ is the exploration rate, and with probability ϵ the agent selects a random action to encourage exploration of the environment. This is typically referred to as an ϵ -greedy policy.

Experience replay drives the learning process in DQN, where the agent stores transitions (s_t, a_t, r_t, s_{t+1}) in a replay buffer $\mathcal{B}$. A minibatch of size m is randomly sampled from the buffer and used to update the Q-values of the main network. The Q-value update rule for the main network is given by the following Bellman equation:

$$Q(s_t, a_t; \theta) \leftarrow Q(s_t, a_t; \theta) + \alpha [r_t + \gamma \max_{a_{t+1}} \hat{Q}(s_{t+1}, a_{t+1}; \hat{\theta}) - Q(s_t, a_t; \theta)]. \quad (11)$$

where $\alpha \in (0, 1)$ is the learning rate, controlling the step size for the update. This rule ensures that the main network's Q-values are gradually adjusted toward the target Q-values as the agent interacts with the environment.

The weights of the target network $\hat{\theta}$ are periodically updated by copying the weights of the main network θ every C steps. This technique of using a target network, which is updated less frequently, helps mitigate the problem of unstable updates that can arise when the Q-network itself is used to generate the target Q-values during training.

To optimize the main network, the DQN minimizes the loss function $L(\theta)$, which is the mean squared error between the current Q-values $Q(s_t, a_t; \theta)$ and the target Q-values $r_t + \gamma \max_{a_{t+1}} \hat{Q}(s_{t+1}, a_{t+1}; \hat{\theta})$:

$$L(\theta) = \mathbb{E} \{ [r_t + \gamma \max_{a_{t+1}} \hat{Q}(s_{t+1}, a_{t+1}; \hat{\theta}) - Q(s_t, a_t; \theta)]^2 \}$$

where the expectation is taken over a minibatch of transitions sampled from the replay buffer. The loss function is minimized via gradient descent, which updates the parameters θ of the main network. As the loss function converges, the Q-values of the main network become stable, and the agent's policy converges to the optimal policy π^* , which selects the action a^* that maximizes the Q-value for each state s :

$$a^* = \pi^*(s) = \text{argmax}_a Q^*(s, a; \theta). \quad (12)$$

In summary, the DQN algorithm integrates several key techniques to stabilize training: experience replay, ϵ -greedy exploration, and the use of target networks. When the loss function $L(\theta)$ converges, the Q-values

accurately approximate the true Q-values, and the agent's policy approaches optimality.

5.3. DQN-SV-RS

Reward shaping is an effective technique that serves the agent in selecting better actions [5]. In this study, we incorporate reward shaping into the DQN algorithm to address dynamic replenishment under seasonal variations, generating a method referred to as DQN-SV-RS.

Our reward-shaping mechanism is initially inspired by the zero-inventory paradigm, where the ideal inventory state is defined as maintaining no inventory while still meeting customer demand [50]. To quantify deviations from this ideal inventory state during period t' , we introduce the concept of the *zero-inventory gap*, which is defined as:

$$P_{t'} = - \left| \sum I_{t'} - d_{t'} \right|. \quad (13)$$

When the remaining inventory exceeds zero, i.e., $\sum I_{t'} - d_{t'} \geq 0$, the inventory is considered excessive. Conversely, when the shortage exceeds zero, i.e., $d_{t'} - \sum I_{t'} \geq 0$, the inventory is considered a stockout. Both scenarios lead to either high holding costs or high stockout costs, deviating from the ideal "zero inventory" state. These negative impacts can be attributed to previous actions of overordering or underordering, which should be penalized *retrospectively* through our reward-shaping mechanism.

Lemma 1. For any previous action $a_t = O_t$ taken during period t , its negative impact on future inventory in period t' is evaluated using a penalty weight $\lambda_{t,t'}$:

$$\lambda_{t,t'} = \begin{cases} \frac{O_t \cdot \mathbb{1}_{t' \in T'_t}}{\sum_{j=1}^{t'} O_j \cdot \mathbb{1}_{t' \in T'_j}}, & \text{if } \sum I_{t'} \geq d_{t'} \\ \frac{(O_{\max} - O_t) \cdot \mathbb{1}_{t' \in T'_t}}{\sum_{j=1}^{t'} (O_{\max} - O_j) \cdot \mathbb{1}_{t' \in T'_j}}, & \text{o.w.} \end{cases}, \quad (14)$$

where $T'_j = \{j + l_j + 1, \dots, j + l_j + k\}$, and $O_{\max} = \max_{j=1,2,\dots,t'} (O_j \cdot \mathbb{1}_{t' \in T'_j})$. Notably, T'_j represents the set of possible periods during which the inventory from action a_j may remain stored. To avoid division by zero, if $O_j = 0$ or $(O_{\max} - O_j) = 0$, we set $O_j = e$ or $(O_{\max} - O_j) = e$, where e is a small positive constant.

Proof. The detailed proof is provided in Appendix B.

We then construct a *zero-inventory penalty* Φ_t that depends solely on the period t and is independent of the state s_t . This is because, after each episode, Φ_t can be calculated via data from the replay buffer indexed by t rather than by s_t . This penalty discounts the negative impact of action a_t on all future inventories in period $t' \in T'_t$, tracing back to period t for evaluation and penalization. This penalty is defined as:

$$\Phi_t = \rho \sum_{t' \in T'_t} \gamma^{t'-t} \lambda_{t,t'} P_{t'}, \quad (15)$$

where $\rho > 0$ is an adjustment penalty factor, a regularization coefficient, used to balance dimensional scales, and where γ is the discount factor in the cumulative discounted reward.

The zero-inventory penalty Φ_t is composed of a zero-inventory gap P_t and a penalty weight $\lambda_{t,t'}$, which together form the core components of the proposed reward-shaping mechanism. In more general forms of reward shaping, a shaped reward function is expressed as:

$$r'_t = r_t + f_t, \quad (16)$$

where the original reward r_t is transformed into a shaped reward r'_t by adding the reward shaping f_t . With these prerequisite above, we construct and prove the proposed reward shaping mechanism, referred to as the *cumulative penalty-based reward shaping function* f_t^p .

Theorem 1. For any state s , action a , and discount factor γ , a cumulative penalty-based reward shaping function f_t^p , which depends solely on period t , is constructed as:

$$f_t^p = \Phi_t - \gamma \Phi_{t+1}, \quad (17)$$

the function f^p guarantees optimal policy invariance (when learning from $M' = (s, a, \gamma, r + f^p)$ rather than from $M = (s, a, \gamma, r)$), i.e., $\pi_M^*(s) = \pi_{M'}^*(s)$. Furthermore, f^p guarantees improved learning performance under M' compared with M , thereby demonstrating its utility.

Proof. The complete proofs of the optimal policy invariance of f^p and its utility are provided in Appendix B.

Although Eq. (17) shows that the reward-shaping function f^p includes Φ_{t+1} (which is related to the next period's action a_{t+1}), it is intuitive to believe that $r'_t = r_t + f_t^p$ for the current period's action a_t is influenced by both Φ_t and Φ_{t+1} . However, it is only Φ_t . A detailed explanation can be found in the proof of utility in Appendix B.

After each episode (i.e., after period T), the DQN-SV-RS agent evaluates the negative impact of its actions on the future inventory by updating its original reward r_t . This update is carried out in a temporary replay buffer $\mathcal{B}'$, where the modified reward in all T time steps is calculated as $r'_t = r_t + f_t^p$. The temporary replay buffer $\mathcal{B}'$ is then merged into the main replay buffer $\mathcal{B}$, which is used for training. This procedure is commonly adopted in related works (e.g., Oroojlooyjadid et al. [20]). By following this procedure, the agent can more effectively learn to choose actions that minimize both future lost sales and net inventory levels, ultimately reducing inventory costs.

Based on the preceding discussion, our DQN-SV-RS dynamic replenishment approach is summarized in Fig. 4. The agent first obtains the state s_t from the IoT system, as depicted in Fig. 1. Using this state, the DQN agent determines the optimal order quantity (action a_t). After executing the action, the agent receives the shaped reward r'_t to evaluate the action. The complete process is outlined in Algorithm 1.

6. Simulation experiments and analysis

In this section, we numerically validate the effectiveness of our proposed dynamic replenishment approach, DQN-SV-RS, through extensive numerical experiments and robustness analysis. In particular, the value of the shaped reward is estimated via an econometric regression model with panel data. Moreover, we conducted an empirical study on real-world data sourced from the Wuxi Chaoyang wholesale market. Most importantly, we substitute the wholesaler with a retailer, demonstrating the generalizability of our DQN-SV-RS approach in the FPSC.

6.1. Experimental settings

6.1.1. Benchmarks

The evaluation compares the performance of our DQN-SV-RS with several benchmarks, including DQN-SV and the original DQN, to highlight the effectiveness of considering seasonal variations and reward shaping. The benchmarks are set up as follows:

- **Predict-then-Optimize (PTO)** is a general dynamic replenishment approach executed in two steps: first, prediction and then, optimization [18]. In our scenario, PTO uses a long short-term memory (LSTM) network, which is well suited for time series with seasonal cycles [51], to predict the future lead time $\hat{l}_{t+1}$ and demand series $\{\hat{d}_i\}_{i=t}^{t+1+\hat{l}_{t+1}}$. The optimal order quantity is then determined by minimizing local inventory costs, as given in Eq. (18). To ensure fairness, the inputs and parameters match those used in our DQN-SV-RS.

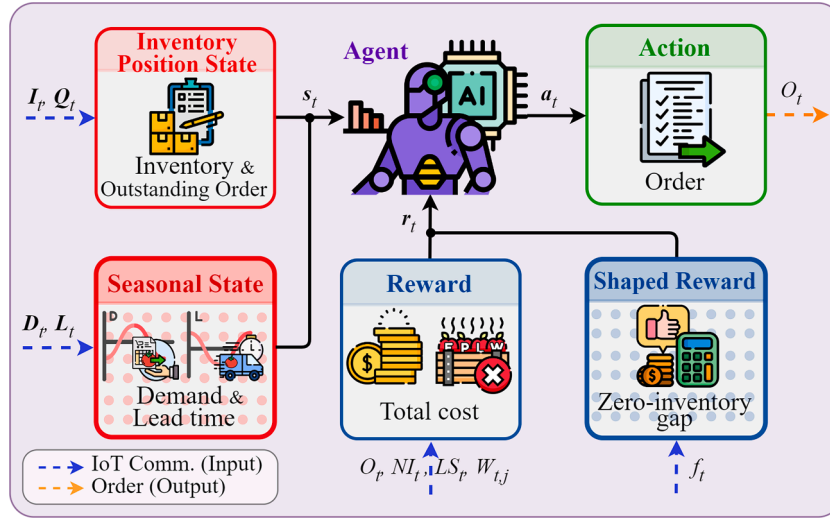


Fig. 4. The MDP Schematic Diagram of Our DQN-SV-RS Approach.

Algorithm 1
DQN-SV-SR Approach.

```

1 Initialize replay buffer  $\mathcal{B}$  to capacity  $N$ 
2 Initialize the main network  $Q$  and target network  $\hat{Q}$  with random weights  $\theta$  and  $\hat{\theta}$ , respectively
3 For episode = 1 :  $M$  do
4   Read the initial state  $si_1$  and seasonal state  $ss_1$  from the IoT cloud server
5   Set  $s_1 = (si_1, ss_1)$ 
6   For  $t = 1 : T$  do
7     Initialize temporary replay buffer  $\mathcal{B}'$  to capacity  $T$ 
8     With probability  $\epsilon$  choose a random action  $a_t$ , otherwise  $a_t = \arg \max_{a_t} Q(s_t, a_t; \theta)$ 
9     Execute action  $a_t$ , and then read state  $s_{t+1} = (si_{t+1}, ss_{t+1})$  and reward  $r_t$  from the IoT cloud server
10    Store transition  $(s_t, a_t, r_t, s_{t+1})$  in temporary replay buffer  $\mathcal{B}'$ 
11    Sample randomly minibatch of transitions  $(s_j, a_j, r'_j, s_{j+1})$  from replay buffer  $\mathcal{B}$ 
12    Set  $y_j = \begin{cases} r'_j, & \text{if } j = T \\ r'_j + \gamma \max_{a_{j+1}} \hat{Q}(s_{j+1}, a_{j+1}; \hat{\theta}), & \text{otherwise} \end{cases}$ 
13    Perform gradient descent step on  $(y_j - Q(s_j, a_j; \theta))^2$  over the network weights  $\theta$ 
14    Every  $C$  iterations, set  $\hat{\theta} = \theta$ 
15  End For
16  Shape the reward  $r'_t = r_t + f_t$  in all  $T$  steps in temporary replay buffer  $\mathcal{B}'$ 
17  Merge temporary replay buffer  $\mathcal{B}'$  into replay buffer  $\mathcal{B}$ 
18 End For

```

$$\begin{aligned}
& \min_{O_{t,t+l_t+1} \geq 0} \sum_{j=t+l_t+1}^{(t+1)+\hat{l}_{t+1}} c_o O_{t,t+l_t+1} + c_h \left[NI_t + O_{t,t+l_t+1} - \sum_{i=t}^j \hat{d}_i \right]^+ \\
& + p_l \left[\sum_{i=t}^j \hat{d}_i - NI_t - O_{t,t+l_t+1} \right]^+
\end{aligned} \quad (18)$$

- **Rolling BSP-low-EW (Rolling-BIE)** is an extension of the BSP-low-EW (BIE), recognized as the optimal perishable replenishment policy [6], as defined in Eq. (19). This benchmark incorporates historical data, along with the predicted lead time $\hat{l}_t$ and demand series

$\{\hat{d}_i\}_{i=t}^{t+\hat{l}_t+1}$ output from the LSTM for seasonal rolling. To ensure a fair comparison, this benchmark aligns with the historical and predicted data used in DQN-SV-RS and other benchmarks, allowing for seasonal feature integration.

$$O_t = \begin{cases} [S_1 - A \cdot IP_t + W_t]^+, & \text{if } IP_t < B; \\ [S_2 - IP_t + W_t]^+, & \text{o.w.} \end{cases}, \quad (19)$$

where $IP_t = \sum_{i=t-k+1}^t \hat{d}_{t,i} + \sum_{i=t+1}^{t+p_{\max}^w+1} R_i$ and $A = 1 - (S_2 - S_1)/B$.

- **End-to-End (E2E)** is a one-step framework that directly generates the order quantity from input features, including time series demand

and lead time [19]. As a state-of-the-art DL approach for inventory management, E2E uses a multi-quantile RNN for demand and a fully connected (FC) network for lead times. For fair comparison, we adopt the E2E network with the same inputs as our DQN-SV-RS and other benchmarks.

- **DQN** has been studied for inventory management [20], so we take it to demonstrate the importance of considering seasonal variations in replenishment. The MDP is parameterized with the inventory position state si_t , action a_t , and reward r_t , which is consistent with DQN-SV-RS. As a control group, this benchmark does not account for seasonal variations.
- **DQN-SV** extends DQN by incorporating seasonal variations, allowing us to evaluate the impact of reward shaping in DQN-SV-RS. The MDP has the parameters of the inventory position state si_t , seasonal state ss_t , action a_t , and reward r_t , which is consistent with DQN-SV-RS for fair comparison.

In summary, these benchmarks, detailed in Table 2, facilitate a comprehensive and fair evaluation of our proposed DQN-SV-RS approach.

6.1.2. Parameter settings

In our environment, the shelf life of fresh produce is set to $k = 3$ (such as 3 days for strawberries). The unit costs are set as follows: ordering $c_o = 3$, inventory holding $c_h = 1$, lost sales penalty $p_l = 5$, and waste penalty $p_w = 7$. Each replenishment period is limited to $T = 28$ days (approximately 1 month), with models trained for 10,000 episodes. Seasonal variations in demand are generated via Eqs. (4) and (5) with the parameters of $d_{\min} = 1$, $d_{\max} = 6$, $l_{\min} = 1$, $l_{\max} = 2$, $l'_{\min} = 1$, $l'_{\max} = 3$. The initial inventory levels are set to 1 and 2 times the average market demand for retailers and wholesalers, respectively. The seasonal state history covers the last $\tau = 10$ periods. Owing to the difficulty in obtaining IoT data for estimating FPLW, we simulate the process of fresh produce degradation via a Weibull distribution, with parameters configured as $a = 0.02$, $b = 1.5$ and $c = 1$ (details in Appendix C).

With respect to the hyperparameters, the input layer size of the neural network is equal to the length of the state. For DQNs, the input layer size is $k + l'_{\max} + 1$ and $k + l'_{\max} + 1 + 4\tau$ for both DQN-SV and DQN-SV-RS, respectively. The output layer of the neural network matches the action space $\mathcal{A} = \{0, 1, \dots, O_{\max}\}$, with a size of $O_{\max} + 1$. Following Oroojlooyjadid et al. [20], we gradually expand the action space to determine the optimal maximum order quantity, ultimately determining that $O_{\max} = 10$ in our experiments. The process of this determination is described in Appendix D. The hidden layers include 3 fully connected layers with 2^5 or 2^6 nodes and ReLU activation. Training uses ϵ -greedy with an exploitation rate increasing to 0.9 over 1000 episodes. A replay buffer of size $N = 5 \times 10^4$ and minibatch size $m = 16$ are used. The loss is computed with MSE and a discount factor $\gamma = 0.9$. Weights are updated via the Adam optimizer with $\alpha = 5 \times 10^{-4}$, and the target network updates every $C = 100 \times 28$ steps. The adjustment penalty factor for reward shaping is set to $\rho = 0.75$ after tuning. A case illustrating key hyperparameter tuning, along with the well-performing hyperparameters used in our experiments, is detailed in Appendix E.

PyTorch is used to build the models, which are trained 5 times with varying seeds. All the experiments are run on Intel® Xeon® Platinum 8362C CPUs and GeForce™ RTX 3090 GPUs. The code is publicly available.³

6.2. Performance comparison

6.2.1. Convergence performance

The trained dynamic replenishment policies are encoded within the

Q-networks of the DQNs. The convergence of these policies serves as a pivotal metric for assessing the efficacy of the model. Fig. 5 depicts the asymptotic convergence of the learning curves for three DQNs, along with the corresponding 95 % CIs shaded. The curves were smoothed every 15 episodes.

Specifically, in Fig. 5(a), each loss curve converges quickly, showing no significant variations after the exploration phase. This indicates that the DQN-based policies stabilize after the tuning optimization and plausible exploration training. The convergences in Fig. 5(b)-(f) reveal that, compared with the other two DQN policies, the DQN-SV-RS policy exhibits desirable properties in terms of stability and consistency in training, as evidenced by its stabilized and converging average value. Moreover, its narrower CI indicates that this policy exhibits reduced variation. Additionally, DQN-SV, with the inclusion of the seasonal state, significantly outperforms the DQN in terms of total cost and net inventory. DQN-SV-RS, which incorporates both the seasonal state and reward shaping, outperforms both the DQN and DQN-SV across all the metrics. This improvement is attributed to the seasonal state and reward shaping, which help the agent select better actions and update a better dynamic replenishment policy throughout the training. Despite both DQN-SV and DQN-SV-RS adopting the seasonal state, the training process for DQN-SV-RS is more stable, owing to the use of a shaped reward.

6.2.2. Performance comparison and analysis

To evaluate the performance of DQN-SV-RS, we compare the results of the trained policies across five different test datasets. These datasets are generated by the same method as the training dataset, with each test dataset containing 1000 episodes. We compute the episode-wide averages of evaluation metrics for each policy, including the total cost, net inventory, lost sales, FPLW, and order quantity. On the basis of the metrics, we calculate the performance gap, which quantifies the improvement of DQN-SV-RS relative to each benchmark, using the following formula:

$$Gap_M^m = \frac{|avg_{DQN-SV-RS}^m - avg_M^m|}{avg_M^m},$$

where avg_M^m denotes the episode average of benchmark M in metric m . The comparison results for the averages and gaps are reported in Table 3. Notably, except for order quantities, lower values for the other averages signify better performance, whereas a larger gap indicates that DQN-SV-RS outperforms the benchmark to a greater extent.

Table 3 demonstrates the overall superior performance of DQN-SV-RS across key metrics, including total cost, net inventory, lost sales, and FPLW, compared with various benchmarks. In terms of total cost, DQN-SV-RS achieves significant improvements, with reductions ranging from 24.0 % (vs. PTO) to 2.8 % (vs. DQN-SV). This highlights its cost-effectiveness among all the tested policies.

The most notable improvement is observed in waste reduction. DQN-SV-RS achieves a remarkable reduction in FPLW, achieving a 90.5 % decrease compared with PTO and an 18.8 % decrease relative to E2E. Furthermore, the FPLW gap decreased from 46.0 % for the DQN to only 6.1 % for DQN-SV, where the DQN does not incorporate a seasonal state while DQN-SV does. These findings underscore the importance of accounting for seasonal variations in further enhancing waste reduction, demonstrating the achievement of DQN-SV-RS in this regard. Additionally, DQN-SV-RS, with the added benefit of reward shaping, further reduces FPLW by 6.1 % compared with DQN-SV, demonstrating the potential of reward shaping in reducing waste.

The reward shaping in DQN-SV-RS could significantly improve net inventory performance. Compared with the DQN-SV and DQN, DQN-SV-RS reduces the net inventory by 8.7 % and 9.3 %, respectively. Compared with the benchmarks, our DQN-SV-RS achieves the largest reduction of 63.3 % (vs. PTO) and the smallest reduction of 6.2 % (vs. Rolling-BIE). Interestingly, PTO and Rolling-BIE represent two opposing strategies: PTO reduces lost sales at the expense of higher inventory

³ All code is available at <https://github.com/ZihaoVistonWang/DQN-SV-RS>.

Table 2
Summary of Benchmarks and our Proposed Approach.

	PTO	Rolling-BIE	E2E	DQN	DQN-SV	DQN-SV-RS (Proposed)
Use IoT data as input?	✓	✓	✓	✓	✓	✓
- Use historical data?	✓	✓	✓	✓	✓	✓
- Use inventory data?	✓	✓	✓	✓	×	✓
- Use seasonal data?	✓	✓	✓	✓	×	✓
What method is used for time series?	LSTM	LSTM & Rolling	MQRNN & FC	FC	FC	FC
Minimize cost objective?	✓	✓	✓	✓	✓	✓
- Cost minimization type?	Local & myopia	Local & myopia	Global & hyperopia	Global & hyperopia	Global & hyperopia	Global & hyperopia
Prediction and optimization type?	Sequential	Sequential	One-step	One-step	One-step	One-step

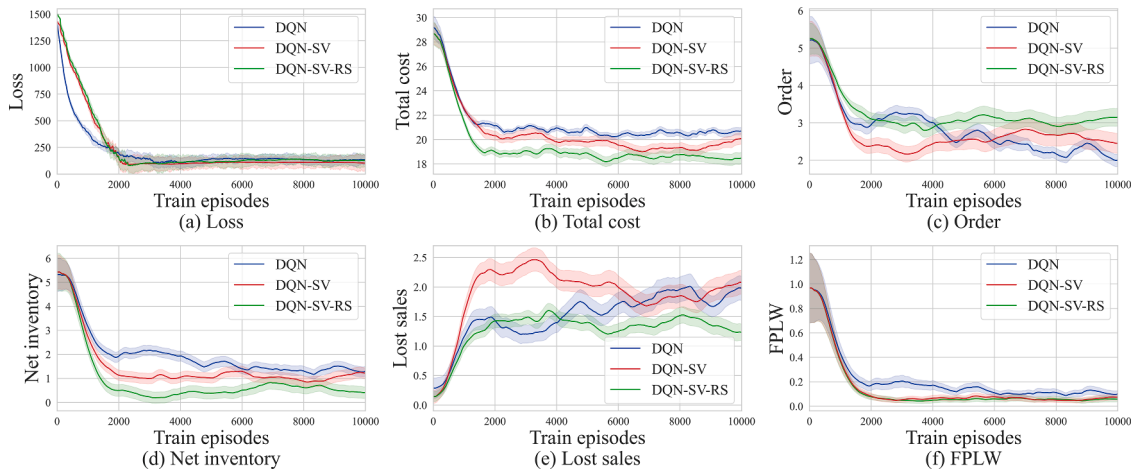


Fig. 5. Learning Curves of the Three DQNs: DQN, DQN-SV, and DQN-SV-RS.

Table 3
Comparisons of DQN-SV-RS with Selected Benchmarks: Average Performance and Gap.

	PTO		Rolling-BIE		E2E		DQN		DQN-SV		DQN-SV-RS
	Average	Gap	Average	Gap	Average	Gap	Average	Gap	Average	Gap	
Total cost	24.678	24.0 %	22.426	16.3 %	20.703	9.4 %	20.506	8.5 %	19.303	2.8 %	18.766
Net inventory	3.308	63.3 %	1.293	6.2 %	1.574	22.9 %	1.336	9.3 %	1.328	8.7 %	1.213
Lost sales	1.756	4.8 %	3.179	47.4 %	2.884	42.0 %	1.975	15.3 %	1.845	9.3 %	1.673
FPLW	0.564	90.5 %	0.073	26.1 %	0.066	18.8 %	0.099	46.0 %	0.057	6.1 %	0.054
Order quantity	2.880	2.0 %	1.607	82.8 %	1.818	61.6 %	2.866	2.5 %	2.784	5.6 %	2.938

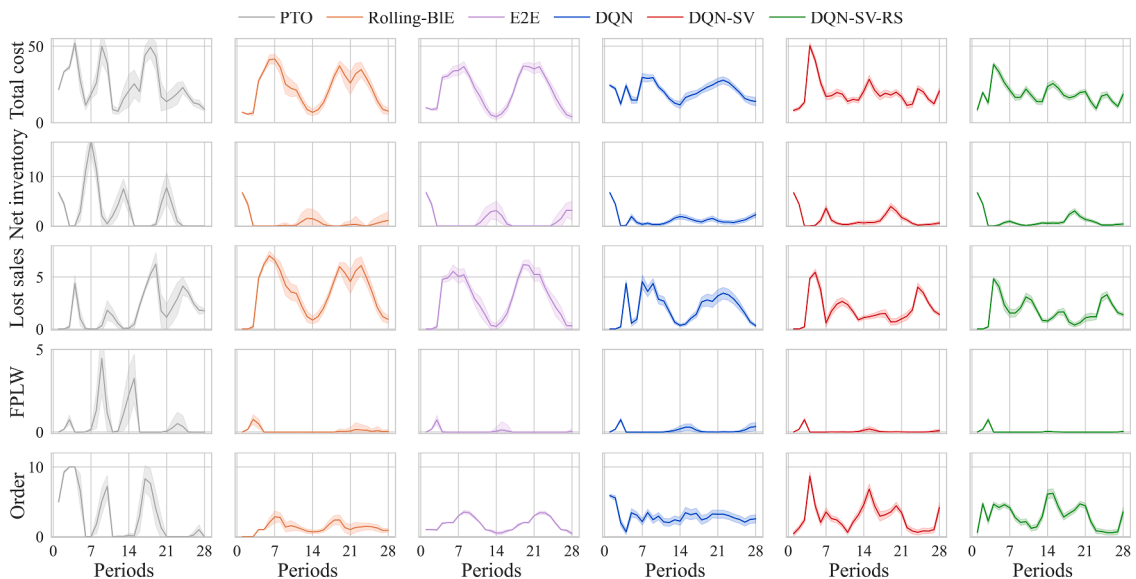


Fig. 6. Variations and Dynamics of Total Cost, Net Inventory, Lost Sales, FPLW, and Order.

holdings, whereas Rolling-BLE minimizes inventory levels at the cost of increased lost sales. DQN-SV-RS, however, strikes a unique balance by achieving lower net inventory while also reducing lost sales by up to 47.4 % compared to Rolling-BLE and by at least 4.8 % compared to PTO. These results testify the effectiveness of the proposed reward-shaping method, which meets the intended design objectives. By optimizing both lost sales and net inventory simultaneously, DQN-SV-RS demonstrates the possibility of achieving "the best of both worlds"—successfully balancing two traditionally conflicting goals.

Interestingly, DQN-SV-RS results in a higher order quantity than the other benchmarks; however, its performance is better in various aspects. This is because reward shaping enables DQN-SV-RS to adjust its ordering actions in alignment with seasonal variations in demand and lead time (as shown in Fig. 6). This results in an effective balance between lost sales and inventory holding, leading to optimal values for these two metrics in Table 3, thereby minimizing the total cost.

In conclusion, combining the results in Table 3, DQN-SV-RS outperforms the other benchmarks across all the performance metrics, which renders DQN-SV-RS the most suitable solution approach for the dynamic replenishment of fresh produce. By incorporating the seasonal state, DQN-SV improves upon the DQN in terms of total cost and FPLW. Moreover, reward shaping enhances DQN-SV-RS by enabling better replenishment decisions, and striking a balance between lost sales and inventory holding.

6.2.3. Visualization analysis of dynamic replenishment

As depicted in Fig. 3, seasonal variations in lead time and demand are both dynamic and stochastic, necessitating a dynamic approach to replenishing fresh produce. Fig. 6 visualizes the dynamic changes in each metric over time. All DQNs, E2E, Rolling-BLE, and PTO dynamically fluctuate. Notably, the CIs for PTO and Rolling-BLE are wider than those of E2E and the three DQNs. This disparity arises because the replenishment strategies of PTO and Rolling-BLE rely on sequential prediction followed by optimization. In contrast, the other four methods employ a more integrated one-step decision-making approach, resulting in greater stability and lower variability across all the metrics.

Owing to space limits, we focus on discussing the replenishment decision in detail, as depicted in the last row of Fig. 6. As shown in Fig. 3, both the demand and lead time exhibit seasonal variations. Periods 7 and 21 represent seasonal demand peaks, whereas periods 0, 14, and 28 correspond to troughs. The seasonal variation in lead time is opposite to that in demand. DQN-SV-SR and DQN-SV, which consider the seasonal state, adjust their order quantities in alignment with the seasonal demand variations. Notably, their order variations as a whole shift approximately three days ahead of the demand variation, indicating that these models utilize IoT-collected seasonal data as critical external information. This enables accurate demand forecasting and proactive replenishment. In contrast, the DQN, which does not account for the seasonal state, maintains a relatively stable ordering pattern. Although the remaining benchmarks also receive the same seasonal data as input and employ time series models to process it (unlike DQN-SV-SR and DQN-SV, which use simple fully connected neural networks), they fail to demonstrate proactive replenishment. This highlights that, under the same scenario of seasonal variability, reinforcement learning's "learning-in-doing" paradigm is more effective for addressing the seasonal demand in fresh produce replenishment than are sequential prediction-optimization or end-to-end deep learning approaches.

Like other benchmarks, DQN-SV-RS cannot perfectly avoid the issue of stockouts during the beginning periods, as large lead times combined with depleted initial inventory cause higher total costs early on. However, DQN-SV-RS demonstrates resilience to early-period stockouts. A comparison of the last two columns in the "Order" row of Fig. 6 shows that the variations in order quantities (peaks and troughs) for DQN-SV-RS are less pronounced than those for DQN-SV. This improvement is attributed to reward shaping, which allows DQN-SV-RS to incorporate seasonal lead time information, reducing order quantities during periods

of shorter lead times and thereby avoiding unnecessarily high inventory levels. This observation is further supported by Fig. 6, which illustrates the variations in order quantities and net inventory for DQN-SV-RS and DQN-SV during periods 3 to 8.

Notably, DQN-SV-RS outperforms the other methods in terms of other aspects, particularly in terms of lost sales and net inventory. For example, as shown in the second row of Fig. 6, the net inventory of DQN-SV-RS consistently remains at a lower level than that of DQN-SV. Additionally, as depicted in the third row of Fig. 6, the lost sales of DQN-SV with a seasonal state are much lower than those of most other benchmarks, but the lost sales of DQN-SV-RS, which incorporates reward shaping, are slightly lower than those of DQN-SV. These findings indicate that the reward shaping we designed successfully achieves the dual objectives of maintaining low inventory levels while reducing lost sales.

6.2.4. Econometric analysis for the value of the reward shaping

In the preceding sections, we examined the superiority of DQN-SV-RS over several benchmarks. Table 3 provides an approximate reflection of the contribution of reward shaping by comparing the average results of DQN-SV-RS and DQN-SV. However, these period-averaged results may be disproportionately skewed due to the influence of specific periods, potentially inflating or diminishing the overall effect. To address this, we first structure the experimental results as panel data and then utilize a two-way fixed effects model as given below:

$$Metric_{it} = \beta_1 is_RS_{it} + \beta_2 X_{it} + \mu_i + \lambda_t + \epsilon_{it}. \quad (20)$$

In the above equation, $Metric_{it}$ represents the logarithms of the performance metrics (e.g., the total cost, net inventory, lost sales, and FPLW) in experiment i at period t ; is_RS_{it} indicates whether a shaped reward is used, with DQN-SV-RS as 1 and DQN-SV as 0. The variable X_{it} includes control variables, including demand and lead time, with β_2 being the estimated coefficient. Additionally, μ_i and λ_t capture the unit and time fixed effects (FEs), respectively, whereas ϵ_{it} is the error term. Notably, setting $Metric_{it}$, the logarithm of the corresponding metric, as the dependent variable allows the regression coefficient β_1 to represent the percentage change in the corresponding metric when is_RS_{it} transitions from 0 to 1. This approach provides a more accurate and clearer estimation of the value of reward shaping.

Table 4 presents the regression results for Eq. (20). Specifically, Columns (1) "Total Cost", (2) "Net Inventory" and (3) "Lost Sales" show that the coefficients for is_RS_{it} are -0.021 , -0.026 and -0.044 , respectively, both of which are significant at the 1 % level. Column (4) "FPLW" reports that a coefficient of -0.012 is significant at the 5 % level. This finding indicates that reward shaping significantly reduces total cost, net inventory, lost sales and FPLW by 2.1 %, 2.6 %, 4.4 % and 0.2 %, respectively. The observed reduction can be attributed to the effectiveness of reward shaping in minimizing total cost by alleviating overstock and stockouts. Furthermore, as shown in the second row of Fig. 6, DQN-SV can maintain nearly zero inventory in the middle and end terms, resulting in a 2.6 % reduction in net inventory when reward

Table 4
Value of the Reward Shaping by Two-way FEs Model.

	(1) Total cost	(2) Net inventory	(3) Lost sales	(4) FPLW
is_RS	-0.021^{***} (0.006)	-0.026^{***} (0.005)	-0.044^{***} (0.005)	0.002^{**} (0.001)
Controls	Yes	Yes	Yes	Yes
Unit FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	56,000	56,000	56,000	56,000
R-squared	0.1523	0.422	0.297	0.381

Note: * $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

shaping is applied. Therefore, reward shaping based on the "zero-inventory" paradigm proves to be effective. Additionally, reward shaping better controls net inventory, thereby reducing the occurrence of excess fresh waste.

However, the values in Table 4, *ceteris paribus*, are all lower than the corresponding results previously reported in Table 3, namely, 1.8 %, 8.7 %, 9.3 %, and 6.1 %. This finding indicates that the contributions of reward shaping, as estimated by the two-way fixed effects model, are more accurate and statistically significant when accounting for time and unit FEs, thus mitigating the influence of specific periods and experiments.

6.3. Empirical study with real business data

To test the performance of our DQN-SV-RS under more practical conditions, we crawled real-world electronic settlement data from the Wuxi Chaoyang Wholesale Market.⁴ This dataset, collected by the market's IoT-enabled agricultural product distribution platform, includes records such as fresh produce names, sales volumes, origins, etc. It provides an indirect means of assessing the model's integration with IoT systems. In this context, sales volume serves as a proxy for demand, whereas the distance from the origin to the market is calculated using the Amap API⁵ and subsequently converted into lead times.

Fig. 7 illustrates the time series of daily sales and distances for four selected fresh produce commodities (i.e., water spinach, Chinese yam, asparagus, and okra), highlighting the bidirectional seasonal variations in demand and lead times. Note that the letter "N" indicates the sample size, while "T" denotes the time window for the observed cycles. Graphically, each of the four commodities exhibits clear seasonal variations with two periodic cycles. Due to the distinct supply and sales characteristics of each fresh produce commodity, the time window for these cycles varies, ranging from the shortest $T = 15$ to the longest $T = 35$. Consequently, the real-world data comprehensively reflects the complexity and authenticity of bidirectional seasonal variations from the periodic and time-window perspectives. Additionally, Fig. 7 includes quantile-quantile (Q-Q) plots, which visually compare the distributions of real-world data and cosine-function-based synthetic data without considering differences in scale. If the sorted quantiles of the two datasets align linearly in the Q-Q plot, their distributions are consistent and follow a normal distribution as we generally assumed in the model. The figure reveals that while the synthetic data is discrete and the real-world data is continuous, leading to horizontal segments in the plot, the data points for both demand and lead time generally align along an upward linear trend, indicating similar distribution patterns. Results for two additional produce commodities and related experiments are presented in Appendix F.

An important challenge when using real business data is how to effectively train and test model performance. Inspired by Oroojlooyjadid et al. [20], we resampled the training and testing datasets to match the quantities used in Section 6.1. Specifically, we applied the moving block bootstrap (MBB) technique for resampling, which is well-suited for time-dependent sequential data as it preserves the autocorrelation structure of the original dataset [52]. More details on how to process the real-world data are provided in Appendix F. These methods not only facilitate model training but also enable fair comparisons with environments using the same parameters as in Section 6.2, thereby offering a more rigorous evaluation of the practical applicability of DQN-SV-RS in real-world scenarios.

Fig. 8 reports the relative improvements of our DQN-SV-RS in comparison with the selected benchmarks based on real-world data. In general, the DQN-SV-RS model demonstrates significant improvements

over the benchmarks across various metrics, particularly in terms of lost sales and FPLW. These findings indicate that implementing dynamic replenishment with DQN-SV-RS in real-world scenarios is promising and beneficial. Even with the variations in real-world environments, our DQN-SV-RS exhibits remarkable efficiency in terms of reducing the lost sales and FPLW as well.

6.4. Robustness checks

To test the robustness and sensitivity of our DQN-SV-RS approach, we conduct four experiments with (1) different intensities of seasonal variations, (2) different unit penalties, and (3) different deterioration rates.

6.4.1. Performance with different intensities of seasonal variations

In this experiment, we consider a nonseasonal scenario and three seasonal scenarios by adjusting the demand and lead time parameters as follows:

- (1) **No** seasonal variations with $d_t \sim N(3, 1)$ and $l_t^w \sim N(2, 1/3)$;
- (2) **Weak** seasonal variations with $d_{\max} = 3$ and $l_{\max}^w = 2$;
- (3) **Moderate** seasonal variations (base setting) with $d_{\max} = 6$ and $l_{\max}^w = 3$; and
- (4) **Strong** seasonal variations with $d_{\max} = 9$ and $l_{\max}^w = 4$.

As depicted in Fig. 9, under no seasonal variations, DQN-SV-RS outperforms most benchmarks across multiple aspects, with its performance improving as seasonal variations transition from weak to strong. These comparative results align well with the preceding analysis, with our DQN-SV-RS remaining as the optimal performance. The absolute differences between DQN-SV-RS and other benchmarks increase as seasonal variations intensify, particularly in reducing lost sales and total costs. This underscores the effectiveness of reward shaping in mitigating shortages and managing inventory holdings, thereby lowering overall costs. In a scenario with no seasonal variation, DQN-SV-RS performs slightly better than DQN-SV and DQN in terms of net inventory and lost sales, indicating that reward shaping remains effective. In summary, as the seasonal variations in demand and lead time intensify, DQN-SV-RS demonstrates an increasing ability to mitigate sales losses and inventory holding, ultimately reducing total costs. Therefore, it is recommended that wholesalers adopt the DQN-SV-RS approach under conditions of seasonal variations.

6.4.2. Cost analysis by unit penalty

The cost factors deserve to be analyzed since they serve as a reward for training DQN agents. Once the ordered product type, storage environment, and monitoring technology are determined, the unit ordering cost c_o and unit holding cost c_h will remain stable over an extended period. However, the unit lost-sale penalty p_l and unit waste penalty p_w are typically set subjectively by managers to increase service levels and reduce FPLW. While these penalties may not directly translate to cash costs, they impact operational performance by affecting goodwill and corporate social responsibility. Therefore, we explore the impacts of the unit lost sales penalty p_l and the unit waste penalty p_w on the total cost.

Figs. 10 and 11 depict the effect on the total cost of varying the unit lost sales penalty p_l from 2 to 8 and the unit waste penalty p_w from 4 to 10. The sensitivity of the total cost is similar across the three DQN approaches, with this effect being more pronounced when the penalty p_l is low and diminishing as the penalty p_l increases. In contrast, both E2E and Rolling-BIE exhibit stronger linear growth in the total cost, whereas PTO shows a quadratic growth pattern, initially rapid and then slowing. We contend that augmenting the penalty p_l alleviates lost sales with more replenishment; nevertheless, it sacrifices the total cost. When the penalty p_l exceeds 4, the number of lost sales decreases progressively, especially in DQN-SV and DQN-SV-RS, and then the effect on total cost

⁴ The data are available on the website of the Key Agricultural Products Market Information Platform (<https://ncpcxx.moa.gov.cn>).

⁵ Amap (<https://lbs.amap.com>).

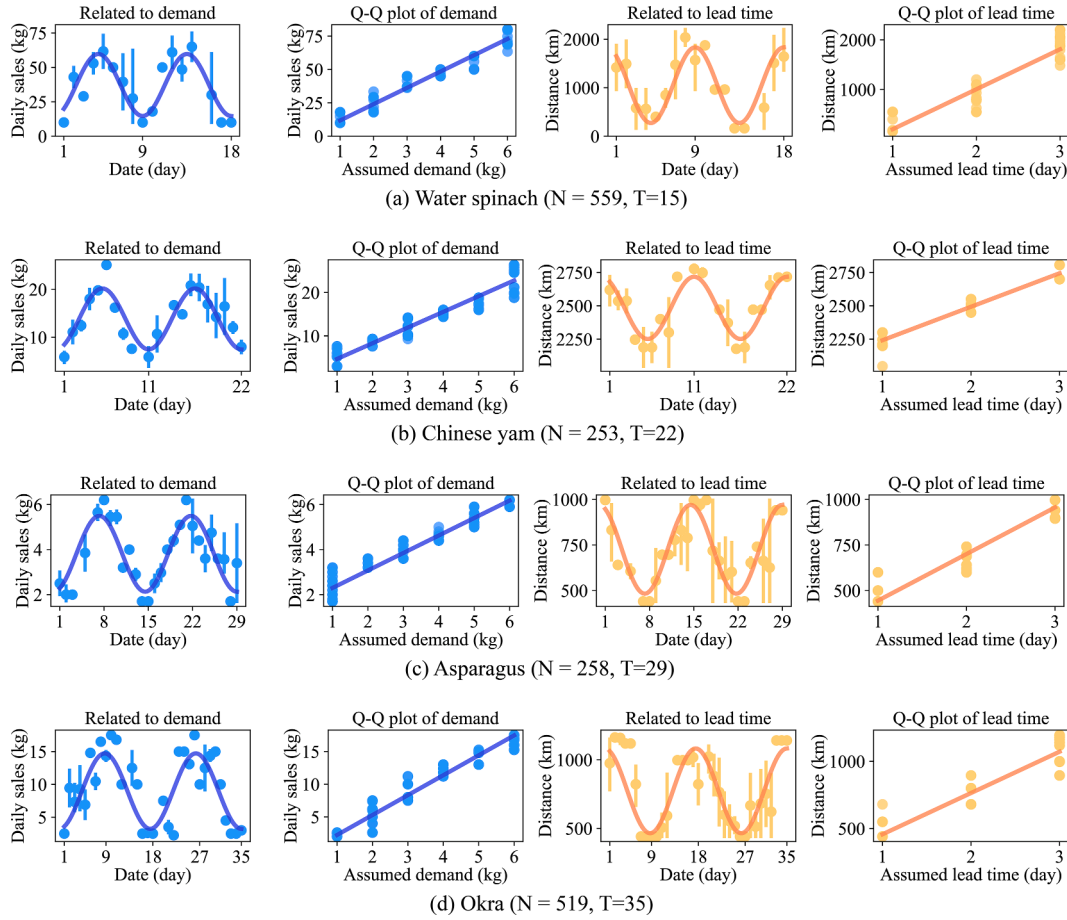


Fig. 7. Seasonal Variations and Q-Q plots in the Real-world Daily Sales (Demand, the Left Two Columns in Blue) and the Distance (Lead Times, the Right Two Columns in Orange) for Four Selected Fresh Produce Commodities.

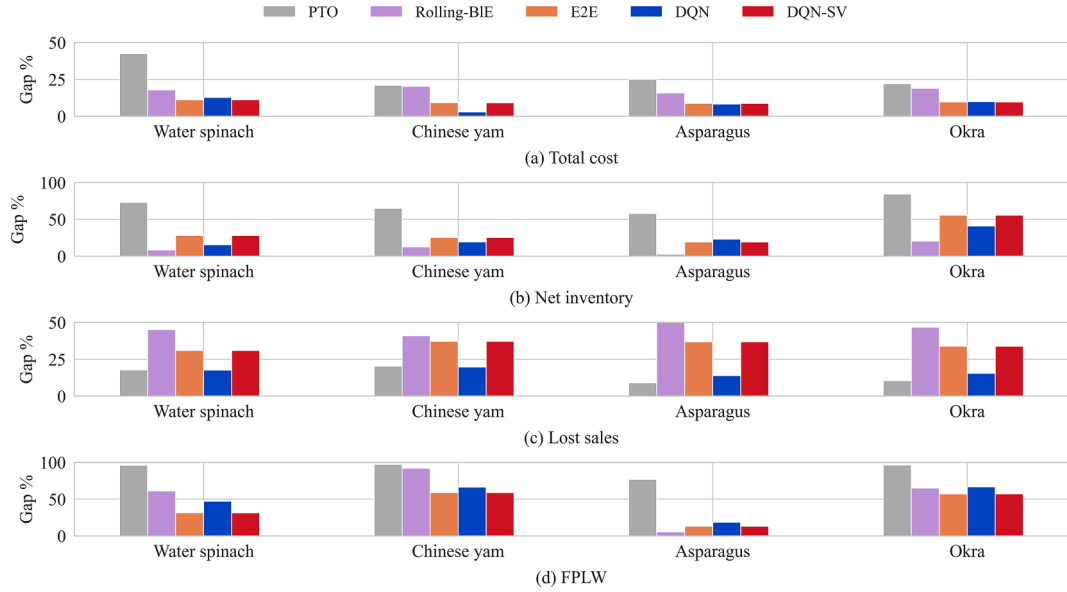


Fig. 8. Relative Improvements of DQN-SV-RS Compared with Benchmarks on Real-world Data.

tends to diminish. This also occurs because the DQN-SV without a shaped reward can also resist the risk of shortage by increasing the penalty p_l . Additionally, as the unit waste penalty p_w increases, both DQN-SV and DQN-SV-RS outperform other benchmarks, with their total costs decreasing due to the reduction in FPLW. In contrast, the total costs of

the other benchmarks increase to varying extents.

Overall, the results support the strong robustness of DQN-SV-RS. Incorporating the seasonal state in DQN-SV-RS and DQN-SV helps minimize lost sales and FPLW, especially under high penalties. With reward shaping, DQN-SV-RS achieves lower total costs. To increase

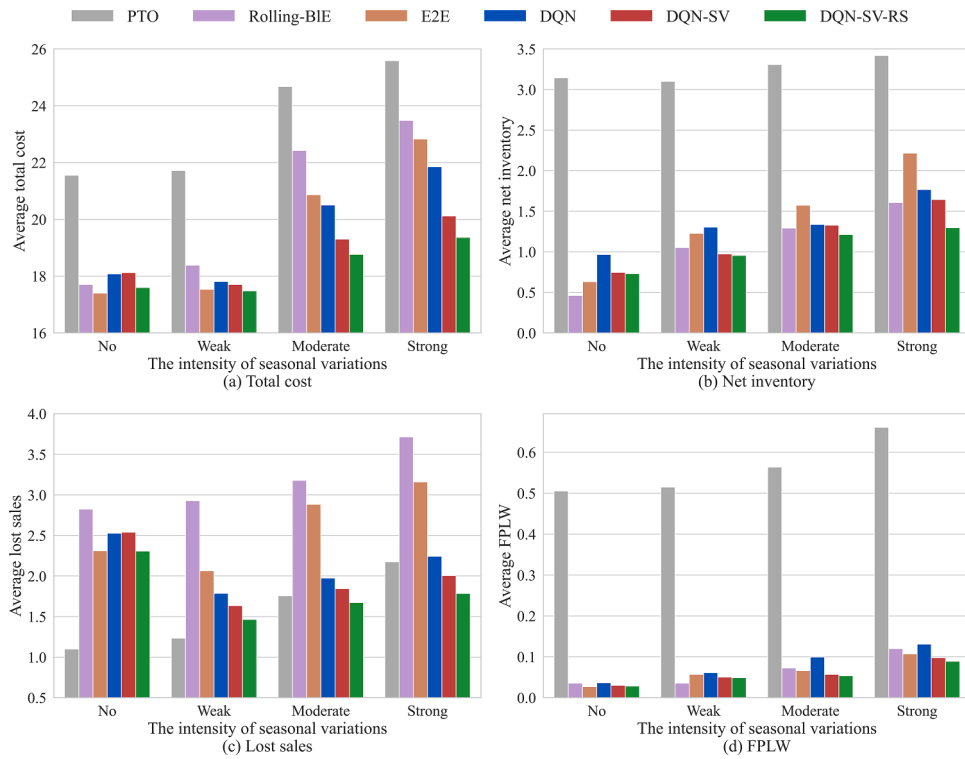


Fig. 9. Robustness check: Comparison across Different Intensities of Seasonal Variations.

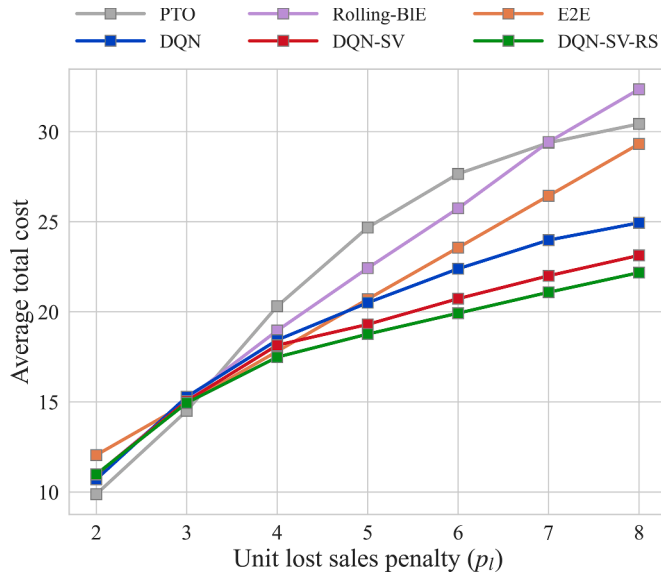


Fig. 10. Comparison of Average Total Cost as the Lost Sales Penalty Varies.

demand fulfillment and reduce food waste, it is advisable to apply higher penalties for lost sales and waste.

6.4.2.1. Effects of deterioration rate on FPLW. The wholesaler agent can learn perishable characteristics from the data collected by the IoT and thereby develop a policy of FPLW reduction. We simulate different fresh produce samples by adjusting the parameters of the Weibull deterioration rate and discuss the effects of perishability on FPLW. In particular, we focus on the shelf life k and the temperature-dependent coefficient b . Here, k reflects whether the fresh produce can be stored for a long time, whereas b reflects how the fresh produce easily deteriorates.

Fig. 12 depicts the average FPLW as the shelf life k increases from 2

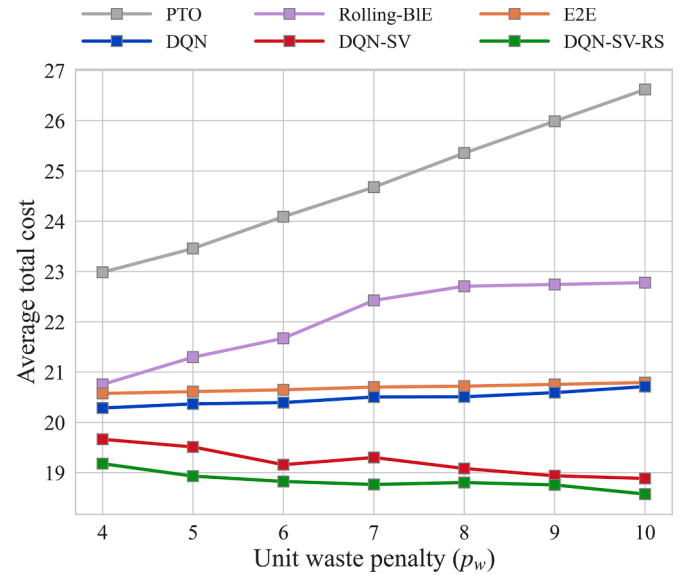


Fig. 11. Comparison of Average Total Cost as the Unit Waste Penalty Varies.

to 5 periods. The average FPLWs for all the approaches initially decrease, but they remain relatively stable after $k = 3$, except for PTO. This stability arises because batches of fresh produce are typically sold out within three days. In summary, in a seasonal context, the replenishment strategy of DQN-SV-RS ensures that fresh produce is sold promptly, minimizing waste due to food expiration. Fig. 13 reported that as the temperature-dependent coefficient b increases from 1 to 2.5, the average FPLW for each model increases to varying degrees, with a diminishing rate of FPLW growth observed for DQNs and E2E at higher values of the coefficient b . In addition, PTO and Rolling-BIE exhibit greater sensitivity to the coefficient b , resulting in more pronounced increases in their FPLWs. The results in Figs. 12 and 13 represent the

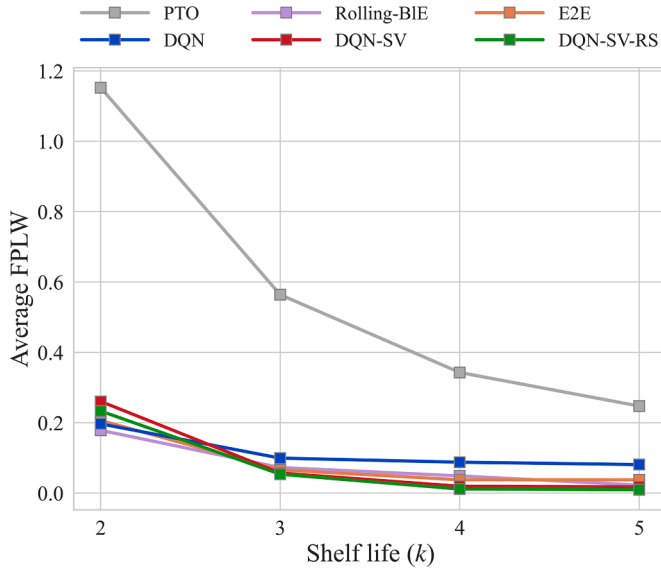


Fig. 12. Comparison of Average FPLW as the Shelf Life Varies.

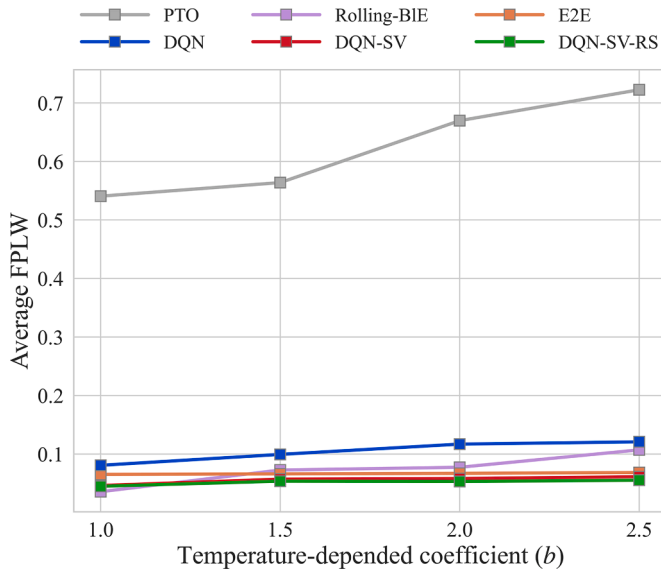


Fig. 13. Comparison of Average FPLW as the Temperature-dependent Coefficient Varies.

stronger robustness of DQN-SV-RS in minimizing FPLW.

6.5. Discussion: retailer's ordering decision compared with that of the wholesaler

To demonstrate the generality of our approach, we present an extended experiment that illustrates the results of a retailer performing

dynamic replenishment of fresh produce. The retailer typically acquires produce from the local wholesaler, so the lead time of the retailer is shorter than that of the wholesaler, and we set it to vary seasonally between 1 and 2. In terms of the supply chain structure, the retailer is located closer to the consumer. In the simulation, we set up the retailer to access the data on consumer demand directly rather than indirectly as the wholesaler does. *Ceteris paribus*, the results of the extended experiments are reported in Table 5.

As exhibited in Table 5, the retailer using DQN-SV-RS also achieves significant improvements across various metrics in the presence of seasonal variations in lead time and demand, which indicates that the proposed DQN-SV-RS has the potential for generalization. Furthermore, compared with the results for wholesalers shown in Table 3, most averages in Table 5 for the retailer are almost less than those in Table 3. This finding indicates that our proposed approach can render better improvements when the lens moves further from a wholesaler (of the upstream) to a retailer (of the downstream) in the FPSC. This can be explained by the bullwhip effect, as the downstream region poses less uncertainty than the upstream region does.

7. Conclusions

This paper studies the replenishment of fresh produce for a wholesaler by considering the seasonal variations in lead time and demand. The proposed system framework integrates the advantage of fully automated IoT monitoring with the ability of DRL to process massive amounts of data to increase the efficiency and performance of dynamic fresh produce replenishment for the wholesaler. With IoT-collected data on seasonal variations in lead time and demand, a dynamic replenishment approach, DQN-SV-RS, is developed based on the DQN approach via reward shaping. The experimental results show that in the context of FPSC, our DQN-SV-RS with seasonal state and reward shaping alleviates lost sales and inventory holding and further optimizes the other metrics of inventory management. In addition, following econometric analysis, the value of the shaped reward lies in mitigating stockouts and overstocking, thereby reducing the total cost. Our proposed DQN-SV-RS demonstrates significant robustness and general applicability, showing superior performance in the application for a retailer.

Based on the experimental results, some managerial insights are provided as follows.

- (1) Considering seasonal variation is critical for the effective replenishment of fresh produce. This enables the FPSC members to resist the bidirectional uncertain risks in lead time and demand, reduce the total cost, and alleviate shortages in the production season and backlogs in the off-production season.
- (2) Fresh produce is significantly saved through our IoT-driven and DQN-based dynamic replenishment, thereby promoting the realization of the sustainable development goal of FPSCs. We call for all stakeholders along the FPSC to adopt our proposed DQN-SV-RS approach for efficient replenishment to reduce wasted inventory, save food, and advance sustainable development.
- (3) Our proposed dynamic replenishment policies can help protect FPSC members from out-of-stock risk and fresh produce waste by setting higher penalties for lost sales and waste.

Table 5

Comparison of average and gap in the retailer.

	PTO		Rolling-BIE		E2E		DQN		DQN-SV		DQN-SV-RS	
	Average	Gap	Average	Gap	Average	Gap	Average	Gap	Average	Gap	Average	Gap
Total cost	21.821	31.1 %	16.617	9.5 %	16.374	8.2 %	16.160	6.9 %	15.704	4.2 %	15.037	
Net inventory	4.023	79.0 %	1.366	38.1 %	1.741	51.4 %	0.902	6.3 %	0.945	10.5 %	0.845	
Lost sales	1.641	9.5 %	2.358	37.1 %	2.168	31.5 %	1.837	19.2 %	1.533	3.2 %	1.484	
FPLW	0.593	92.2 %	0.073	36.8 %	0.052	12.2 %	0.085	45.9 %	0.050	8.8 %	0.046	
Order	3.482	38.2 %	1.250	72.0 %	1.260	70.7 %	1.827	17.7 %	2.248	4.3 %	2.150	

- (4) To put our concept into practice, we propose a framework that enables the DQN agent to process copious amounts of data collected from IoT scenarios for dynamic replenishment of fresh produce, which expands the application of new-generation information technology in FPSC management.

We acknowledge that several extensions can be made for future research. For example, integrating blockchain technology—such as decentralization and smart contracts [53,54]—along with multi-agent DRL technology could enhance the collaborative dynamic replenishment of fresh produce across multiple entities. Additionally, using actual IoT data related to fresh produce degradation (rather than relying on simulation data) and applying advanced techniques, such as the quality degradation prediction model [55], could further increase FPLW reduction and promote the sustainable development of FPSCs.

CRedit authorship contribution statement

Zihao Wang: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Data curation, Conceptualization. **Wenlong Wang:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tianjun Liu:** Validation, Software, Methodology, Conceptualization. **Jasmine Chang:** Writing – review & editing, Visualization, Validation, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Jim Shi:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

All authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China (grant number 72102185), Chinese Universities Scientific Fund (grant number 2452021113, 2452021171), and the National Social Science Fund of China (grant number 22VRC152). The fifth author is grateful to the United States Department of Agriculture (USDA, award numbers AM21TMATRD00C003 and 22-TMTSD-NJ-0003), as well as the Leir Research Institute (LRI) at NJIT.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.omega.2025.103299](https://doi.org/10.1016/j.omega.2025.103299).

Data availability

Data will be made available on request.

References

- [1] World Economic Forum. How technology can help address challenges in agriculture. 2022. Retrieved June 8, 2023 from, <https://www.weforum.org/agenda/2022/03/how-technology-can-help-address-challenges-in-agriculture/>.
- [2] Dong L. Toward resilient agriculture value chains: challenges and opportunities. *Prod Oper Manag* 2021;30(3):666–75.
- [3] Liu HY, Zhang JL, Zhou C, Ru YH. Optimal purchase and inventory retrieval policies for perishable seasonal agricultural products. *Omega-Int J Manag Sci* 2018; 79:133–45.
- [4] Boute RN, Gijsbrechts J, van Jaarsveld W, Vanvuchelen N. Deep reinforcement learning for inventory control: a roadmap. *Eur J Oper Res* 2022;298(2):401–12.
- [5] De Moor BJ, Gijsbrechts J, Boute RN. Reward shaping to improve the performance of deep reinforcement learning in perishable inventory management. *Eur J Oper Res* 2022;301(2):535–45.
- [6] Haijema R, Minner S. Improved ordering of perishables: the value of stock-age information. *Int J Prod Econ* 2019;209:316–24.
- [7] Hou XY, Haijema R, Liu DC. Display, disposal, and order policies for fresh produce with a back storage at a wholesale market. *Comput Ind Eng* 2017;111:18–28.
- [8] Bu J, Gong X, Chao X. Asymptotic optimality of base-stock policies for perishable inventory systems. *Manage Sci* 2023;69(2):846–64.
- [9] Khalifa FB, Kouki C, Safra I, Jemai Z. Analysis and optimisation of periodic inventory models for perishable items with a general lifetime. *Int J Prod Res* 2022; 61(21):7483–501.
- [10] Jing FY, Chao XR. A dynamic lot size model with perishable inventory and stockout. *Omega-Int J Manag Sci* 2021;103:102421.
- [11] Kumar G, Bera S, Samanta G, Maiti M. Optimal profit in two-level trade credit EOQ model with default risk and reminder cost under finite time horizon having time-dependent demand and deterioration. *Rairo-Oper Res* 2024;58(5):3895–921.
- [12] Xie Y, Tai AH, Ching WK, Kuo YH, Song N. Joint inspection and inventory control for deteriorating items with time-dependent demand and deteriorating rate. *Ann Oper Res* 2021;300(1):225–65.
- [13] Li L, Tang O, Zhou W, Fan T. Backroom effect on perishable inventory management with IoT information. *Int J Prod Res* 2021;61(12):4157–79.
- [14] Alvarez A, Cordeau JF, Jans R, Munari P, Morabito R. Inventory routing under stochastic supply and demand. *Omega-Int J Manag Sci* 2021;102:102304.
- [15] Qiu YZ, Qiao J, Pardalos PM. Optimal production, replenishment, delivery, routing and inventory management policies for products with perishable inventory. *Omega-Int J Manag Sci* 2019;82:193–204.
- [16] Qiao X, Wang Z, Chen HX. Joint ordering and markdown policy for short lifetime products with competitive price- and freshness-based demand. *IEEE Trans Automat Sci Eng* 2021;18(4):1956–68.
- [17] Mnih V, Kavukcuoglu K, Silver D, Rusu AA, Veness J, Bellemare MG, Graves A, Riedmiller M, Fidjeland AK, Ostrovski G, Petersen S, Beattie C, Sadik A, Antonoglou I, King H, Kumaran D, Wierstra D, Legg S, Hassabis D. Human-level control through deep reinforcement learning. *Nature* 2015;518(7540):529–33.
- [18] Elmachtoub AN, Grigas P. Smart “predict, then optimize. *Manage Sci* 2022;68(1): 9–26.
- [19] Qi M, Shi YY, Qi YZ, Ma CX, Yuan R, Wu D, Shen ZJ. A practical end-to-end inventory management model with deep learning. *Manage Sci* 2023;69(2):759–73.
- [20] Oroojlooyjadid A, Nazari M, Snyder LV, Takáč M. A deep Q-network for the beer game: deep reinforcement learning for inventory optimization. *Manuf Serv Oper Manag* 2022;24(1):285–304.
- [21] Bai R, Kendall G. A model for fresh produce shelf-space allocation and inventory management with freshness-condition-dependent demand. *INFORMS J Comput* 2008;20(1):78–85.
- [22] Qin YY, Wang JJ, Wei CM. Joint pricing and inventory control for fresh produce and foods with quality and physical quantity deteriorating simultaneously. *Int J Prod Econ* 2014;152:42–8.
- [23] Macías-López A, Cárdenas-Barrón LE, Peimbert-García RE, Mandal B, Sana SS. An inventory model for perishable items with price-, stock-, and time-dependent demand rate considering shelf-life and nonlinear holding costs. *Math Probl Eng* 2021;2021:1–36.
- [24] Sivashankari CK, Ramachandran L. Inventory models with integrated time dependent demands for deteriorating items - in third and fourth order equations. *Rairo-Oper Res* 2021;55(5):2883–905.
- [25] Arnade C, Pick D, Gehlhar M. Testing and incorporating seasonal structures into demand models for fruit. *Agric Econ* 2005;33(3):527–32.
- [26] Udenio M, Vatamidou E, Fransoo JC. Exponential smoothing forecasts: taming the bullwhip effect when demand is seasonal. *Int J Prod Res* 2022;61(6):1796–813.
- [27] Hansen O, Transchel S, Friedrich H. Replenishment strategies for lost sales inventory systems of perishables under demand and lead time uncertainty. *Eur J Oper Res* 2023;308(2):661–75.
- [28] Li S, He Y, Salling M. Strategic rationing and freshness keeping of perishable products under transportation disruptions and demand learning. *Complex Intell Syst* 2022;8(6):4513–27.
- [29] Riezebos J, Zhu SX. Inventory control with seasonality of lead times. *Omega-Int J Manag Sci* 2020;92:102162.
- [30] Alibabaei K, Gaspar PD, Assunção E, Alirezazadeh S, Lima TM. Irrigation optimization with a deep reinforcement learning model: case study on a site in Portugal. *Agric Water Manag* 2022;263:107480.
- [31] Din A, Ismail MY, Shah BB, Babar M, Ali F, Baig SU. A deep reinforcement learning-based multi-agent area coverage control for smart agriculture. *Comput Electric Eng* 2022;101:108089.
- [32] Gijsbrechts J, Boute RN, Van Mieghem JA, Zhang DJ. Can deep reinforcement learning improve inventory management? Performance on lost sales, dual-sourcing, and multi-echelon problems. *Manuf Serv Oper Manag* 2022;24(3): 1349–68.
- [33] Yang C, Feng Y, Whinston A. Dynamic pricing and information disclosure for fresh produce: an artificial intelligence approach. *Prod Oper Manag* 2022;31(1):155–71.
- [34] Mondal S, Wijewardena KP, Karuppuswami S, Kriti N, Kumar D, Chahal P. Blockchain inspired RFID-based information architecture for food supply chain. *IEEE Internet Things J* 2019;6(3):5803–13.
- [35] Zhang LH, Li T, Fan TJ. Radio-frequency identification (RFID) adoption with inventory misplacement under retail competition. *Eur J Oper Res* 2018;270(3): 1028–43.

- [36] Alizadeh-Sani M, Mohammadian E, Rhim JW, Jafari SM. pH-sensitive (halochromic) smart packaging films based on natural food colorants for the monitoring of food quality and safety. *Trends Food Sci Technol* 2020;105:93–144.
- [37] Herbon A, Ceder A. Monitoring perishable inventory using quality status and predicting automatic devices under various stochastic environmental scenarios. *J Food Eng* 2018;223:236–47.
- [38] Hassini E, Ben-Daya M, Bahroun Z. Modeling the impact of IoT technology on food supply chain operations. *Ann Oper Res* 2023.
- [39] Özbilge A, Hassini E, Parlar M. Optimal pricing and donation policy for fresh goods. *Eur J Oper Res* 2024;312(1):198–210.
- [40] Tufano A, Zuidwijk R, Van Dalen J. The development of data-driven logistic platforms for barge transportation network under incomplete data. *Omega-Int J Manag Sci* 2023;114:102746.
- [41] Ben-Daya M, Hassini E, Bahroun Z. Internet of things and supply chain management: a literature review. *Int J Prod Res* 2017;57(15–16):4719–42.
- [42] Wang YY, Liu J, Yang ZH, Qu ZH, Yu N, Xiang W, Ye BL. A protocol stack for large-scale RFID systems: mitigating reader and tag collisions. *IEEE Internet Things J* 2024;11(19):30455–68.
- [43] Vanvuchelen N, Gijsbrechts J, Boute R. Use of proximal policy optimization for the joint replenishment problem. *Comput Ind* 2020;119:103239.
- [44] Disney SM, Maltz A, Wang X, Warburton RDH. Inventory management for stochastic lead times with order crossovers. *Eur J Oper Res* 2016;248(2):473–86.
- [45] Priore P, Ponte B, Rosillo R, de la Fuente D. Applying machine learning to the dynamic selection of replenishment policies in fast-changing supply chain environments. *Int J Prod Res* 2019;57(11):3663–77.
- [46] USDA. By 2019, blueberry production had expanded both geographically and seasonally. 2020. Retrieved December 20, 2022 from, <https://www.ers.usda.gov/d/ata-products/chart-gallery/gallery/chart-detail/?chartId=99987>.
- [47] Cox BD, Whichelow MJ, Prevost AT. Seasonal consumption of salad vegetables and fresh fruit in relation to the development of cardiovascular disease and cancer. *Public Health Nutr* 2000;3(1):19–29.
- [48] Spence C. Explaining seasonal patterns of food consumption. *Int J Gastron Food Sci* 2021;24:100332.
- [49] Chan CK, Zhou Y, Wong KH. A dynamic equilibrium model of the oligopolistic closed-loop supply chain network under uncertain and time-dependent demands. *Transp Res Part E-Logistic Transp Rev* 2018;118:325–54.
- [50] Obermaier R, Donhauser A. Zero inventory and firm performance: a management paradigm revisited. *Int J Prod Res* 2012;50(16):4543–55.
- [51] Bandara K, Bergmeir C, Hewamalage H. LSTM-MSNet: leveraging forecasts on sets of related time series with multiple seasonal patterns. *IEEE Trans Neural Netw Learn Syst* 2021;32(4):1586–99.
- [52] Künsch HR. The jackknife and the bootstrap for general stationary observations. *Annal Stat* 1989;17(3):1217–41.
- [53] Chang A, El-Rayes N, Shi J. Blockchain technology for supply chain management: a comprehensive review. *FinTech* 2022;1(2):191–205.
- [54] Chang J, Katehakis MN, Shi J, Yan ZP. Blockchain-empowered newsvendor optimization. *Int J Prod Econ* 2021;238:108144.
- [55] Xu XL, Guo WG, Rodgers MD. A real-time decision support framework to mitigate degradation in perishable supply chains. *Comput Ind Eng* 2020;150.